



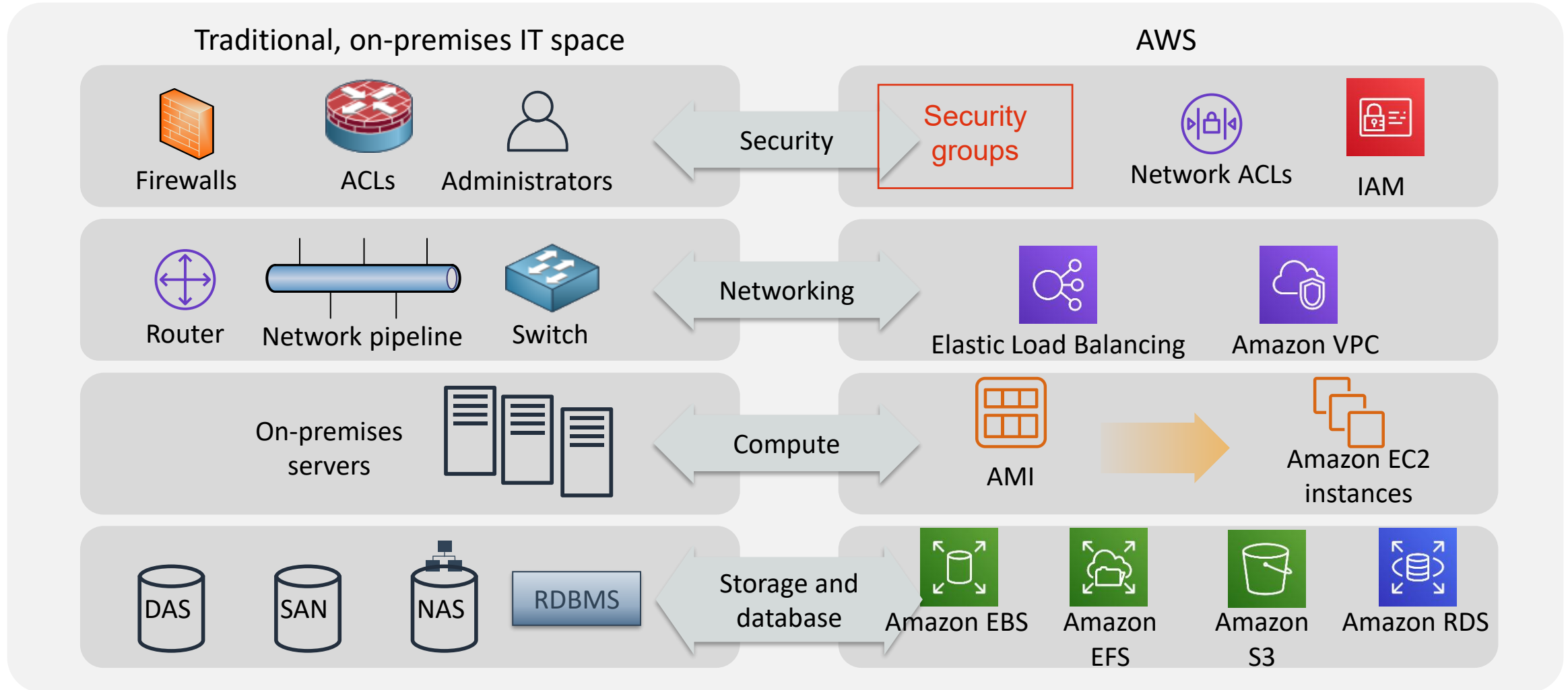
IT332/CS446 Cloud Computing

3 Understanding EBS

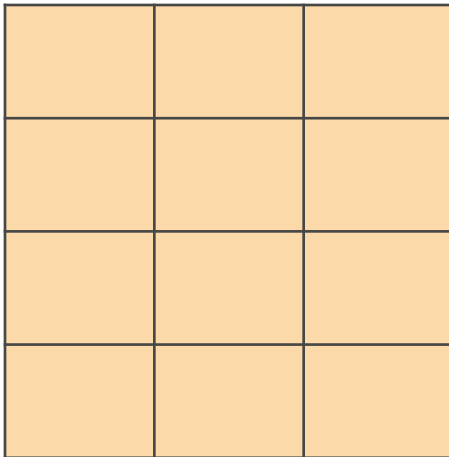
Information Technology
School of Information Technology and Innovation
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CISSP, CISM, CISA, GSSP-JAVA, ITILv3 Expert, IRCA-ISMS, IRCA-BCMS, RHCE

Migrate to AWS Cloud Computing



Amazon Elastic Block Store (EBS)



Block storage

- Block-level storage
 - Individual storage volumes
 - Attach them to an Amazon EC2 instance
 - Volumes are automatically replicated within its [Availability Zone](#)
- Snapshots
 - Backup data on disk with Point-in-Time
- Encryption
 - No additional cost
- Elasticity
 - Increase capacity

Amazon Elastic Block Store (EBS)

Solid State Drives (SSD)		Hard Disk Drives (HDD)	
General Purpose	Provisioned IOPS	Throughput-Optimized	Cold
This type is recommended for most workloads	Critical business applications that require sustained IOPS performance, or more than 16,000 IOPS or 250 MiB/second of throughput per volume	Streaming workloads that require consistent, fast throughput at a low price	Throughput-oriented storage for large volumes of data that is infrequently accessed
System boot volumes	Large database workloads	Big data	Scenarios where the lowest storage cost is important
Virtual desktops		Data warehouses	It cannot be a boot volume
Low-latency interactive applications		Log processing	
Development and test environments		It cannot be a boot volume	

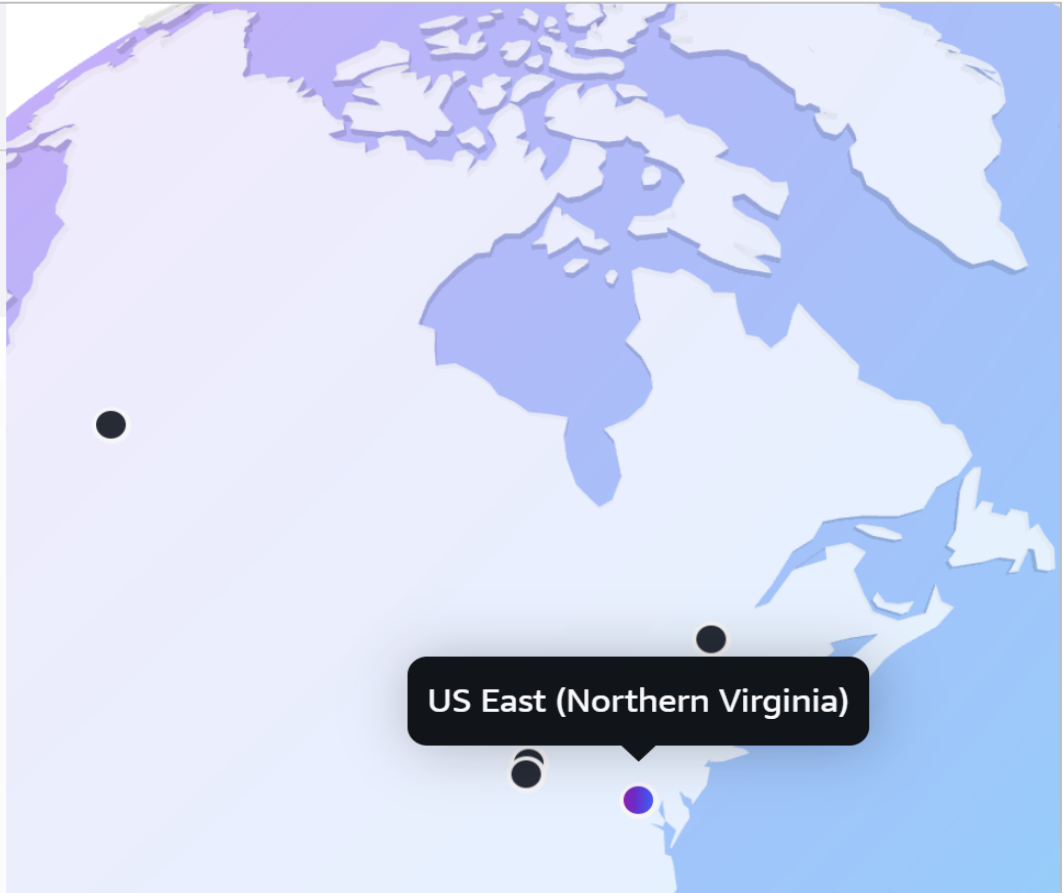
AWS Regions and Availability Zones

- https://aws.amazon.com/about-aws/global-infrastructure/regions_az/

North America

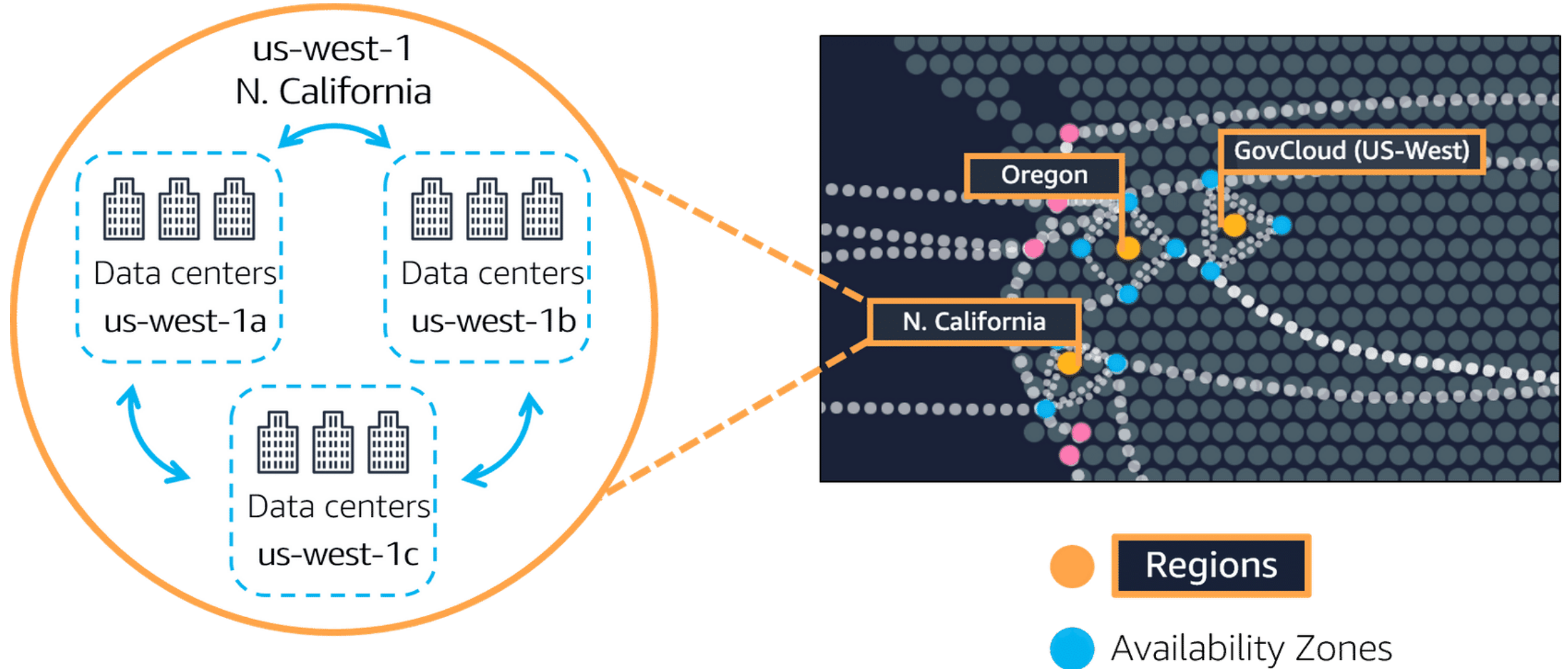
Geographic Regions 9 —

- AWS GovCloud (US-west)
- Canada (Central)
- Canada West (Calgary)
- Mexico (Central)
- US West (Northern California)
- **US East (Northern Virginia)**
Availability Zones: 6 Launched: 2006



The image shows a screenshot of the AWS website's 'Regions and Availability Zones' page for North America. On the left, a list of regions is displayed under the heading 'Geographic Regions'. The 'US East (Northern Virginia)' region is selected and highlighted in black, showing it has 6 availability zones and was launched in 2006. To the right, a map of North America is shown with several black dots representing other AWS regions. A callout box points to the purple dot representing the US East (Northern Virginia) region.

AWS Regions and Availability Zones



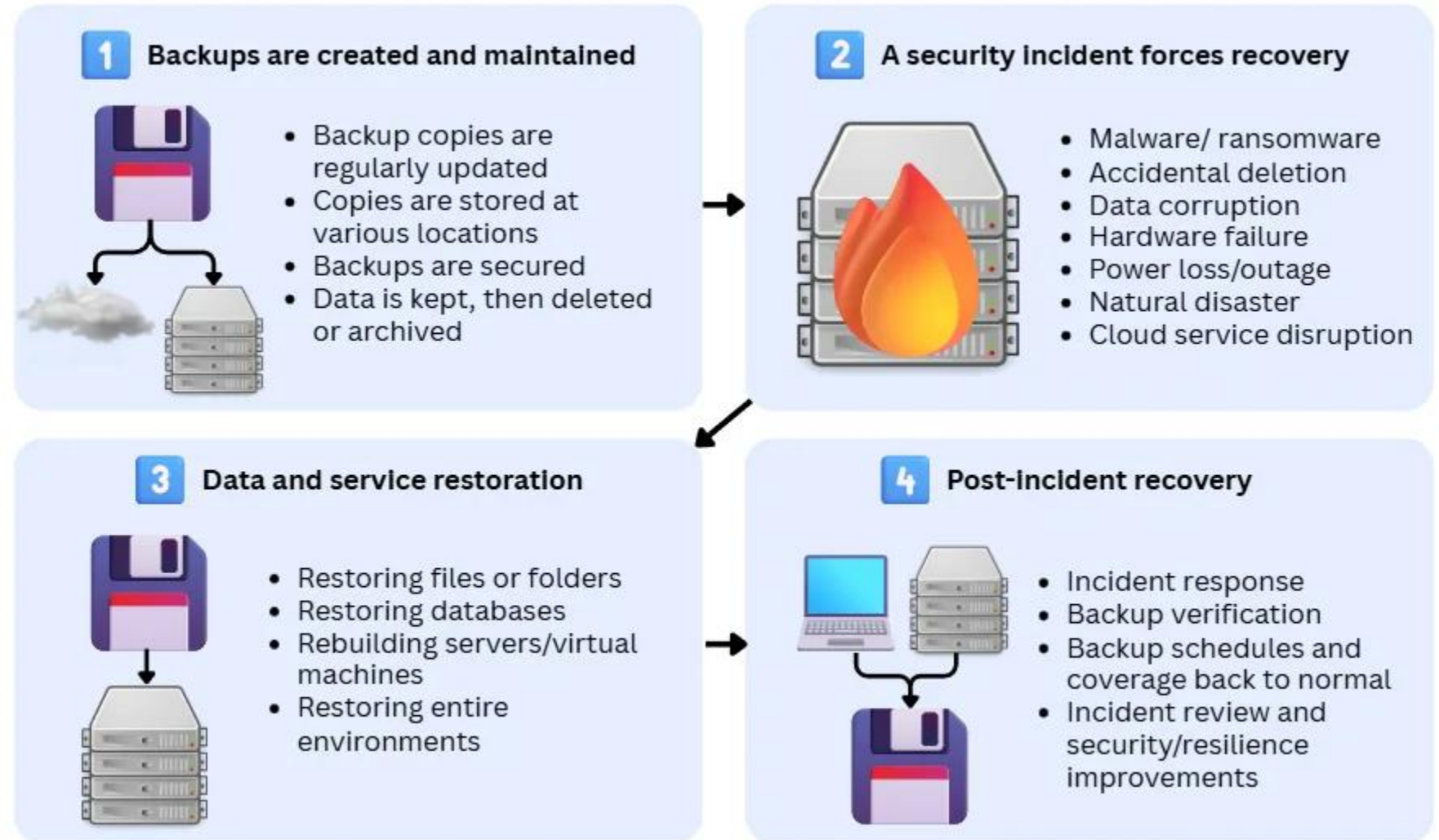
Understanding EBS Lab



- Provision EC2 instance and Web Server
- Manage EC2 Elasticity
 - Resize CPU and Ram
 - Resize Disk
- Create and Attach New EBS Volume
- Manage Web Data
- Back up and Restore Web Data with Snapshot
 - Life saver in any incident cases

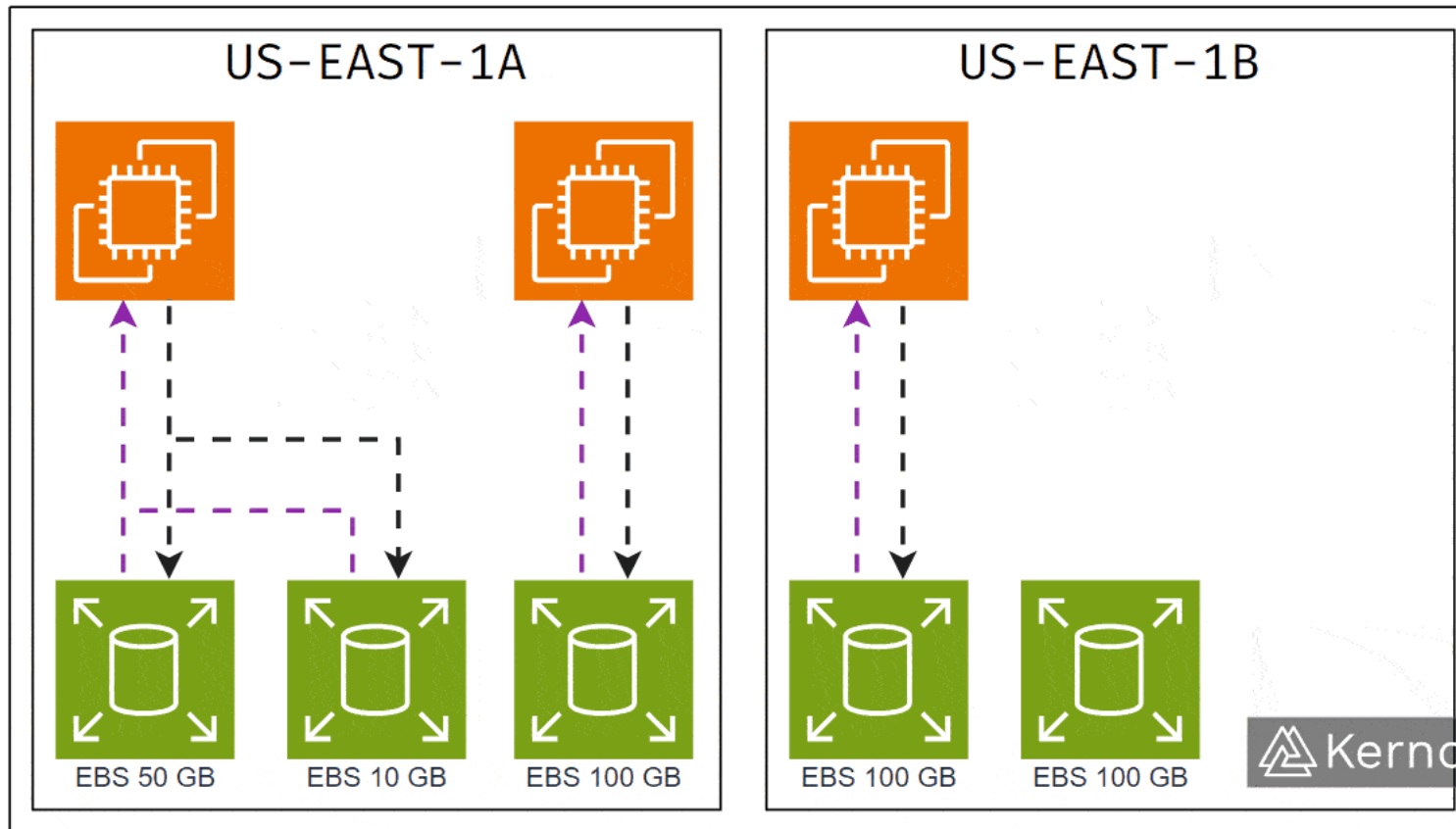
Understanding EBS Lab

- Business data backup is one of the core Cybersecurity process
- Creating and retaining independent, recoverable copies of any Valuable Assets
- Restored following an incident e.g. Ransomware attacks



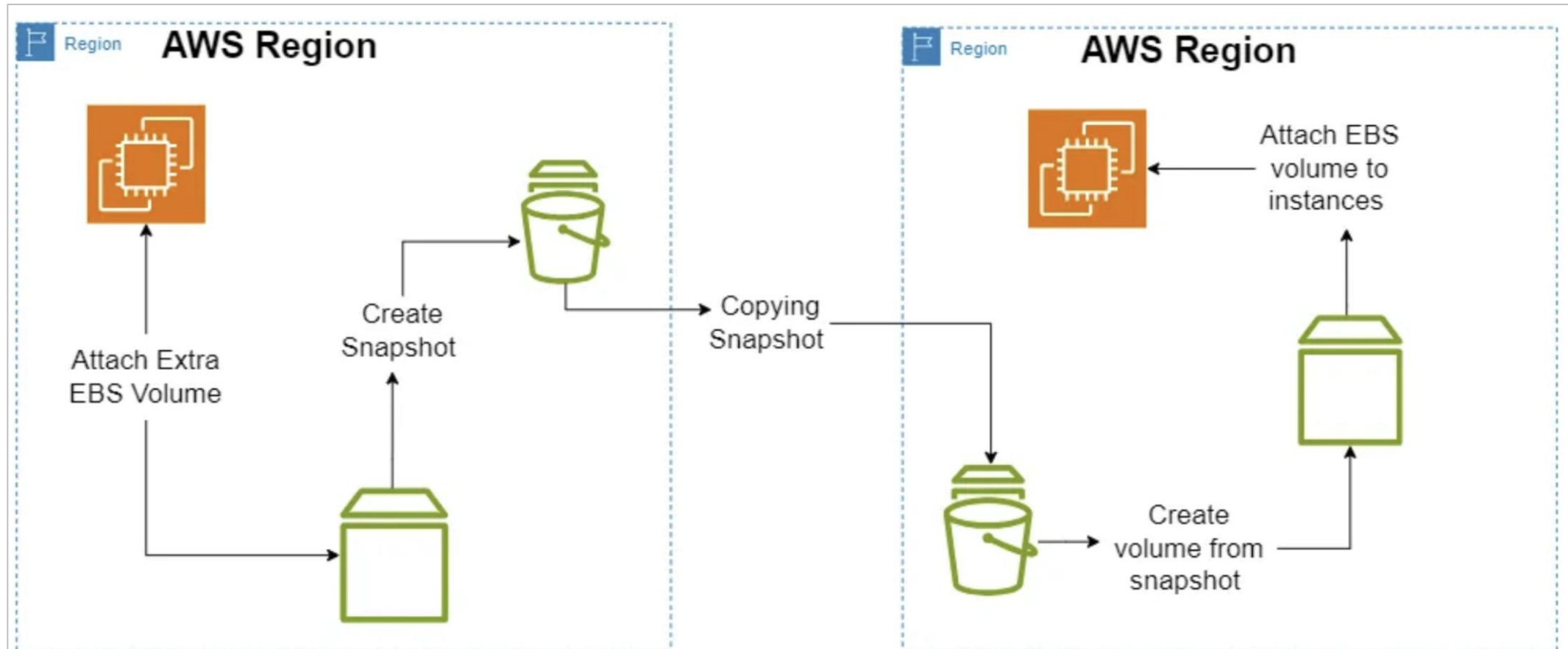
Understanding EBS Lab

- Attach new Disk = Same Region / Availability Zone



Understanding EBS Lab

- Restore to Another Region = **Snapshot Migration**



Tip: Help on Hands-on Lab



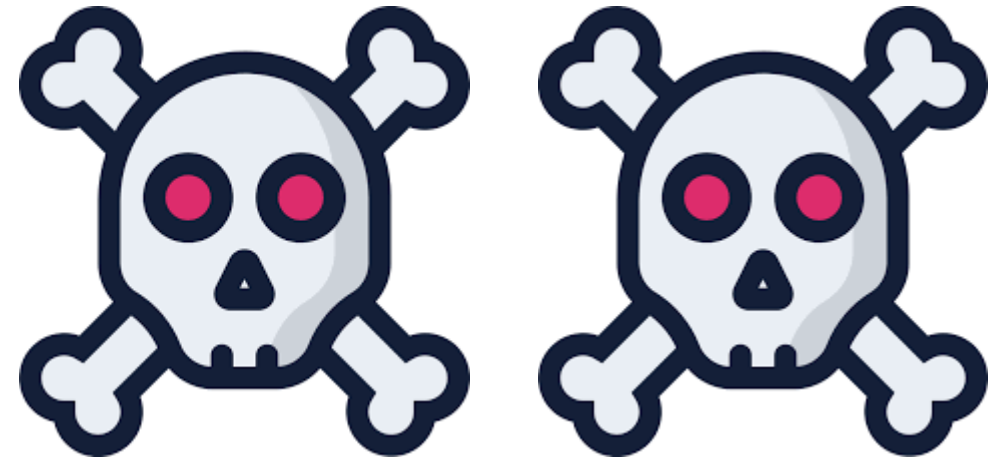
- Practice makes perfect. Focus in completion of the lab
- Self Helping
 - Review All Steps before Performing
 - Error is expected, and good. Just try it again
 - Feel free to redo all steps again
- Group Helping
 - Do peer or group reviewing
 - Check with team first before posting or asking
 - Who done, help your friend in your group
 - Group check but individual grading
- Class Helping
 - Uploaded teaching materials after the class
 - Google classroom comments or Line chat is always available, feel free to post there

Caution: Copy and Paste from Slide

- Be careful for characters substitution when Copy and Paste from PDF to your Command Line
- Below characters, fixed by delete and type yourself

```
$ echo "This is good day" >> a.txt | tee -a
```

“ is NOT ”
– is NOT –



Access AWS Lab

- Search: [aws academy student login](#)

aws academy student login

- Look for Untitled



Instructure

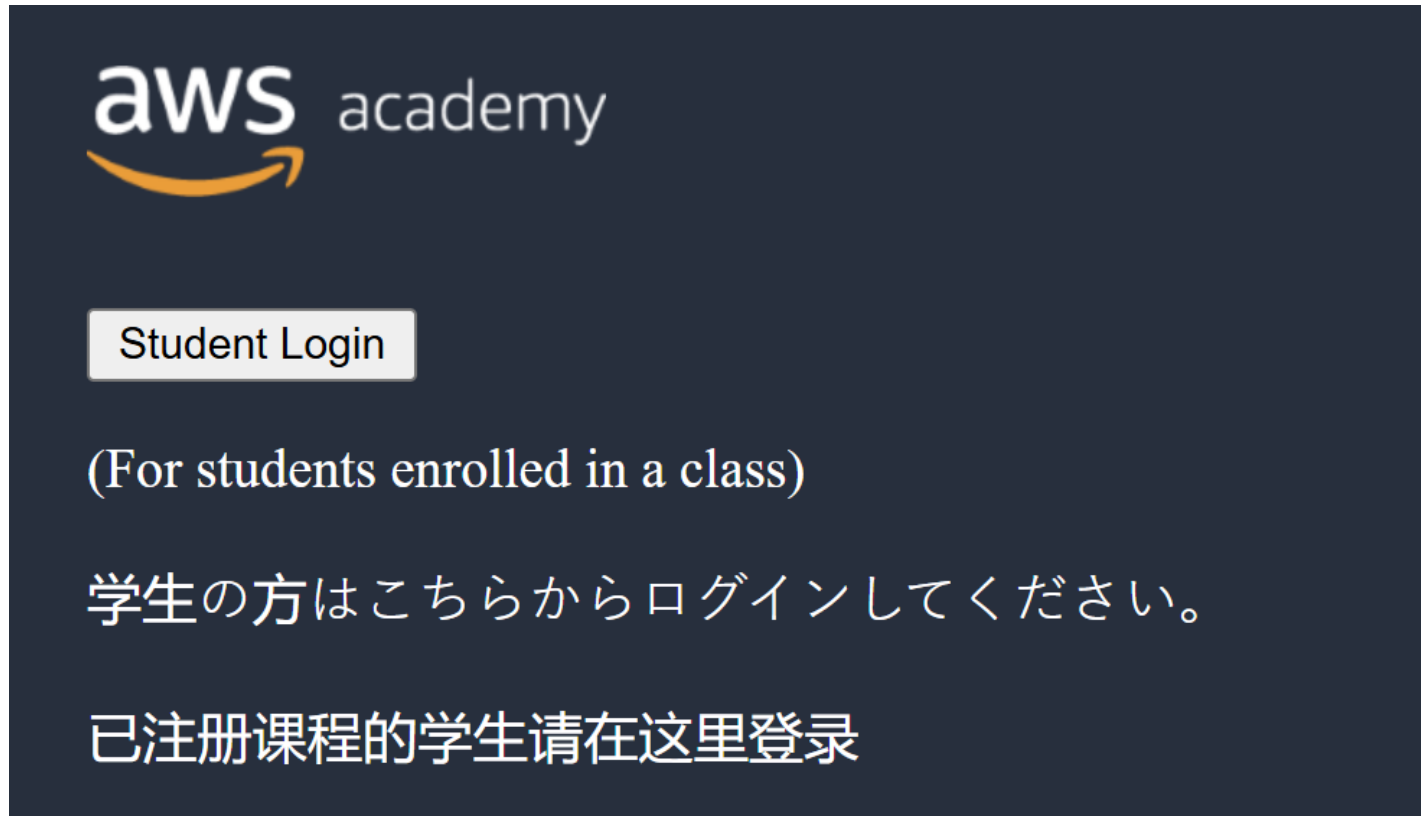
<https://awsacademy.instructure.com> · [Translate this page](#) ⋮

Untitled

学生の方はこちらからログインしてください。 已注册课程的学生请在这里登录. **Educator Login.** (For educators who have **access** to the **AWS Academy Portal**). [Read more](#)

Access AWS Lab

- Choose Student Login



Access AWS Lab

- Login with bumail.net
- OTP in your email



Welcome to Canvas

Email *

Password *

☐ Remember me

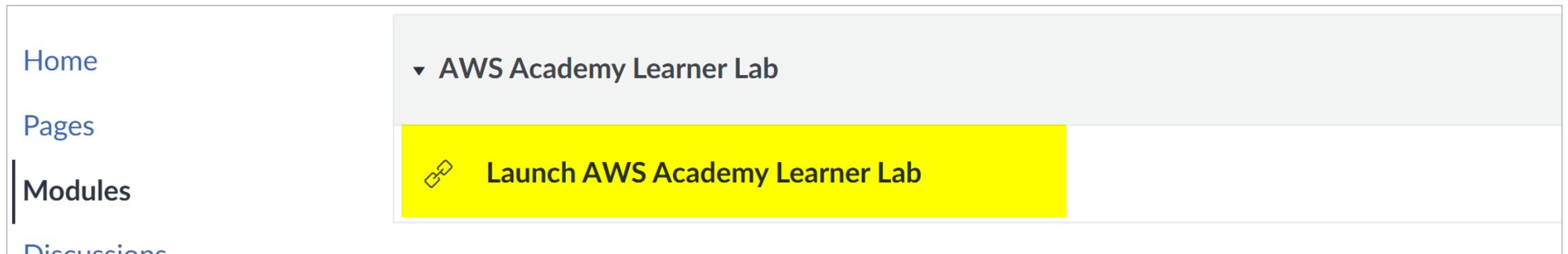
[Log In](#)

[Forgot password?](#)

[Trouble logging in?](#)

Access AWS Lab

- Click **Modules** Tab on the left panel
- Click **Launch AWS Academy Learner Lab**



Access AWS Lab

- Click **Start Lab**

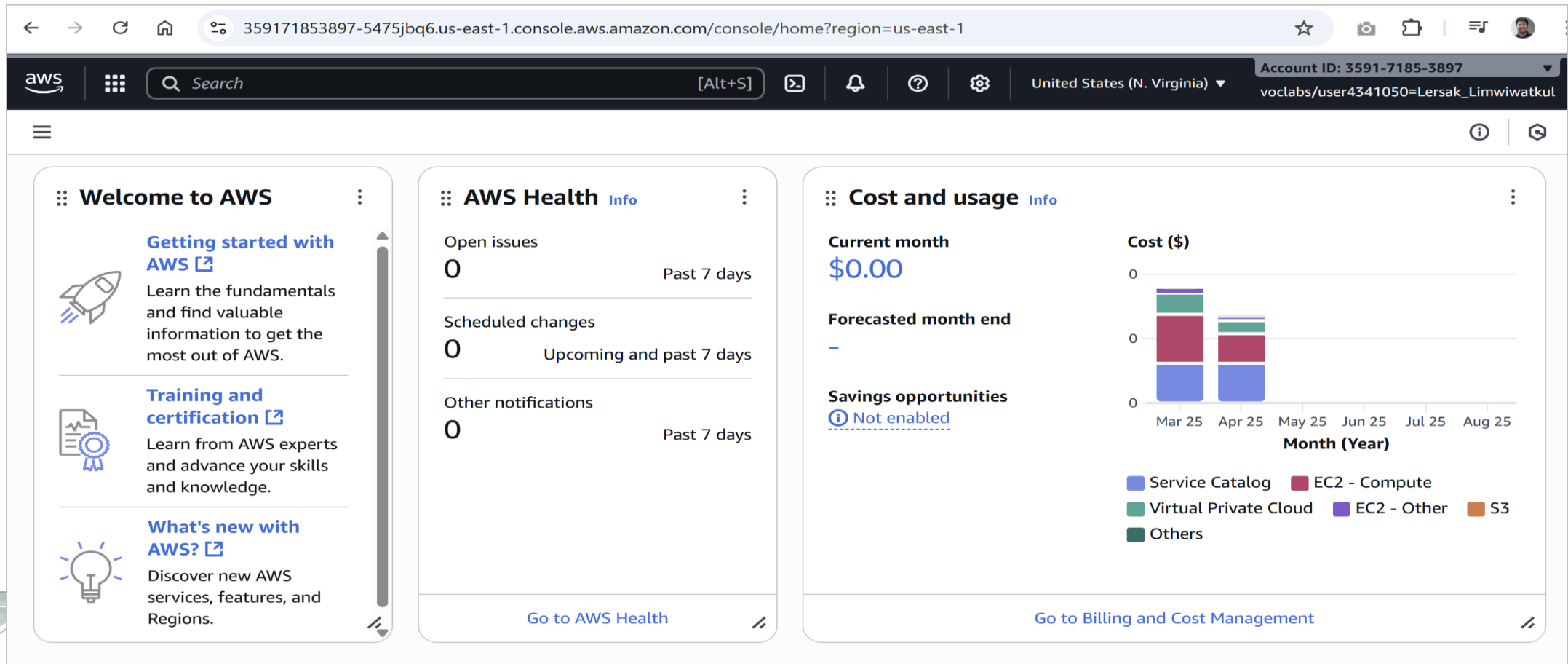


- Time elapsed for Lab
 - Start Lab – Create resources for using within Lab
 - Click again for extending Lab Time
 - End Lab – Terminate all the resources to preserve Credits
 - AWS Details – Lab Details and Access Key Information
- Wait until Lab Status **Show Green**



Access AWS Console

- Click Green AWS
- Open New Tab for AWS Console



The screenshot shows the AWS Management Console interface. The top navigation bar includes the AWS logo, a search bar, and various utility icons. The main content area is divided into three columns:

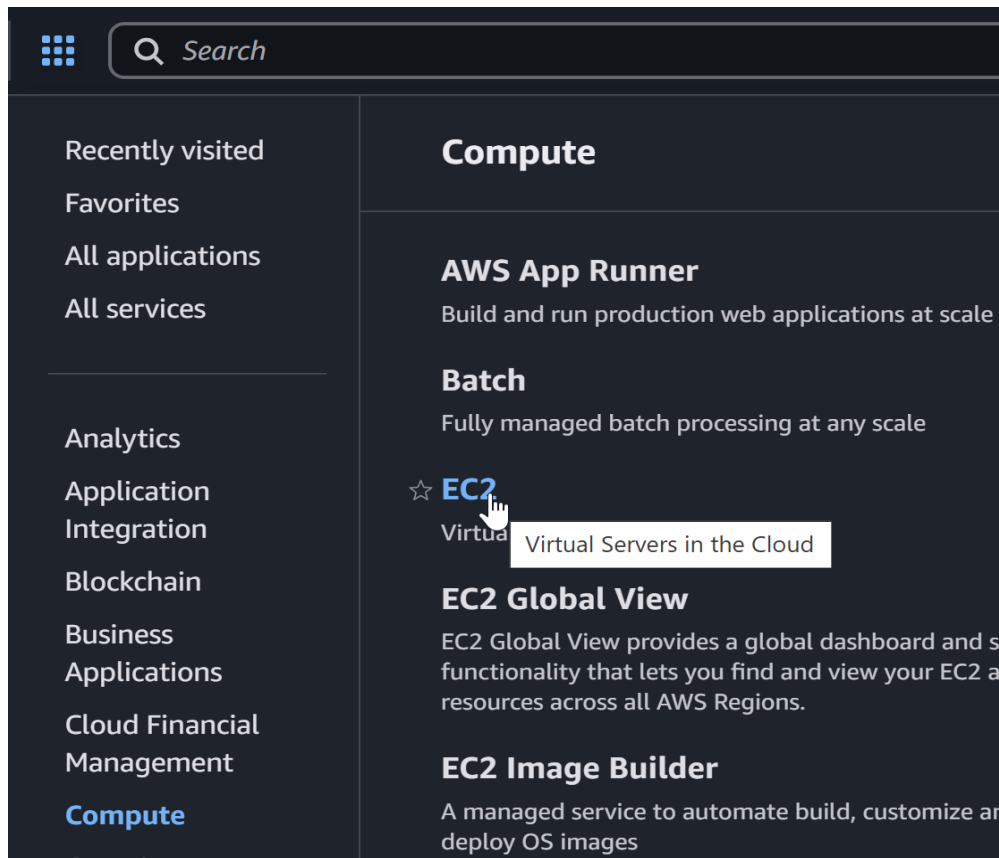
- Welcome to AWS:** Contains links for "Getting started with AWS", "Training and certification", and "What's new with AWS?".
- AWS Health:** Displays "Open issues" (0), "Scheduled changes" (0), and "Other notifications" (0). A "Go to AWS Health" link is at the bottom.
- Cost and usage:** Shows "Current month" cost as \$0.00 and "Forecasted month end" as -. It includes a bar chart for "Cost (\$)" by "Month (Year)" from March to August. The chart shows costs for Service Catalog, EC2 - Compute, Virtual Private Cloud, EC2 - Other, and S3. A "Go to Billing and Cost Management" link is at the bottom.

An aerial photograph of a lush green landscape featuring terraced rice fields. A light-colored, winding river or path cuts through the terraces, which are arranged in a series of concentric, undulating patterns. The fields are filled with water, reflecting the sky and creating a shimmering effect. The overall scene is a harmonious blend of natural beauty and human agricultural engineering.

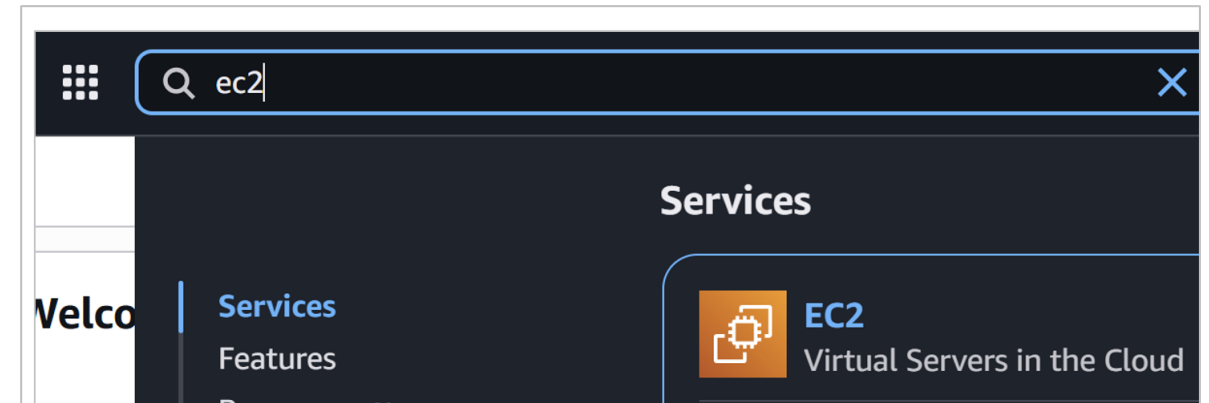
Provision EC2 instance and Web Server

EC2 Console

- Compute > EC2



- Search > EC2



EC2 Console

EC2

<

Dashboard

EC2 Global View

Events

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

▼ Images

AMIs

AMI Catalog

Resources

EC2 Global View

You are using the following Amazon EC2 resources in the United States (N. Virginia) Region:

Instances (running)	1	Auto Scaling Groups	0
Capacity Reservations	0	Dedicated Hosts	0
Elastic IPs	0	Instances	1
Key pairs	1	Load balancers	0
Placement groups	0	Security groups	4
Snapshots	0	Volumes	1

Launch instance

To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.

Service health

AWS Health Dashboard

Region

Account attributes

Default VPC

vpc-02e944721edc43851

Settings

Data protection and security

Allowed AMIs

Zones

EC2 Serial Console

Default credit specification

EC2 console preferences

Explore AWS

Save up to 90% on EC2 with Spot Instances

Optimize price-performance by combining EC2 purchase options in a single EC2 ASG.

Learn more

EC2 Console

- Review Current Instance
- EC2 → Instances
 - Refresh
 - Instance state
 - Actions

EC2 > Instances

EC2

Dashboard

EC2 Global View

Events

▼ Instances

Instances

Instance Types

Launch Templates

Instances (1) Info

Refresh

Connect

Instance state ▼

Actions ▼

Launch instances ▼

Find Instance by attribute or tag (case-sensitive)

All states ▼

< 1 >

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☐	Bastion Host	i-00b7b9e37950d21c0	Running	t2.micro	2/2 checks passed	View alarms

Remove existing Instances (If Any)

- Select any instance
- Instance state → Terminate (delete) instance

The screenshot displays the AWS Management Console's 'Instances' page. At the top, there's a header with 'Instances (1/1)' and an 'Info' link. Below this is a search bar with the placeholder text 'Find Instance by attribute or tag (case-sensitive)'. A table lists the instances, with columns for selection, Name, Instance ID, and Instance state. One instance, 'joe111111' with ID 'i-06bfb6611c64d8550', is shown in a 'Running' state. To the right of the table, the 'Actions' dropdown menu is open, showing several options: 'Stop instance', 'Start instance', 'Reboot instance', 'Hibernate instance', and 'Terminate (delete) instance'. A mouse cursor is pointing at the 'Terminate (delete) instance' option.

	Name	Instance ID	Instance state
☒	joe111111	i-06bfb6611c64d8550	Running

Create an new instance - Name

- Name and tags
- Your Student ID as Machine Name
–joe111111

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started with the simple steps below.

Name and tags [Info](#)

Name

[Add additional tags](#)

Create an Instance - AMI

- Application and OS Images
- Amazon Machine Image (AMI)
 - Ready to use OS Images
- Select **Ubuntu 24.04 LTS**

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

🔍 Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux
aws

macOS
Mac

Ubuntu
ubuntu®

Windows
Microsoft

Red Hat
Red Hat

SUSE Linux
SUSE


[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

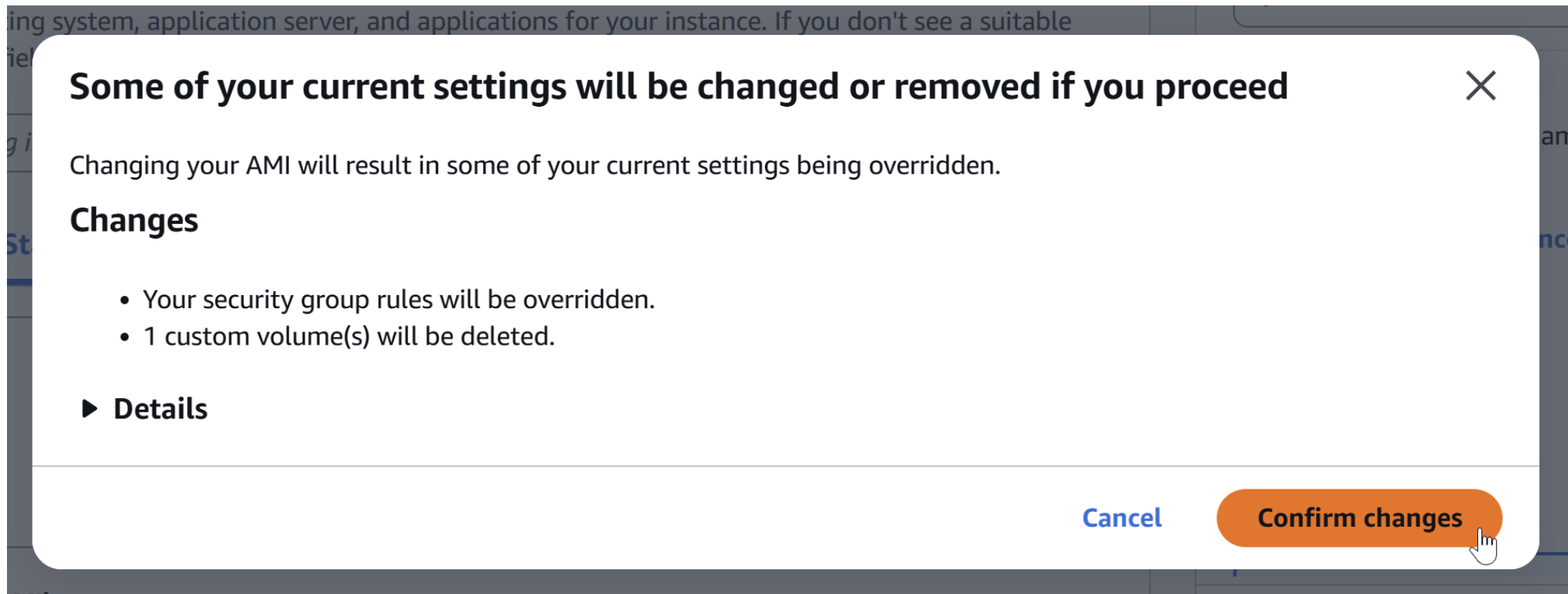
Amazon Machine Image (AMI)

Ubuntu Server **24.04 LTS** (HVM), SSD Volume Type
ami-0360c520857e3138f (64-bit (x86)) / ami-026fccd88446aa0bf (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

Create an Instance – Choose AMI

- If Pop-up
- Click **Confirm changes**



Create an Instance – Instance type

- Instance type
- Select **t3.micro**

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro

Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour

On-Demand SUSE base pricing: 0.0104 USD per Hour

On-Demand Linux base pricing: 0.0104 USD per Hour

On-Demand RHEL base pricing: 0.0392 USD per Hour

On-Demand Windows base pricing: 0.0196 USD per Hour

☒ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

Create an Instance – Key pair

- SSH Key Pair to Login to our instance
- Select **vockey**

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

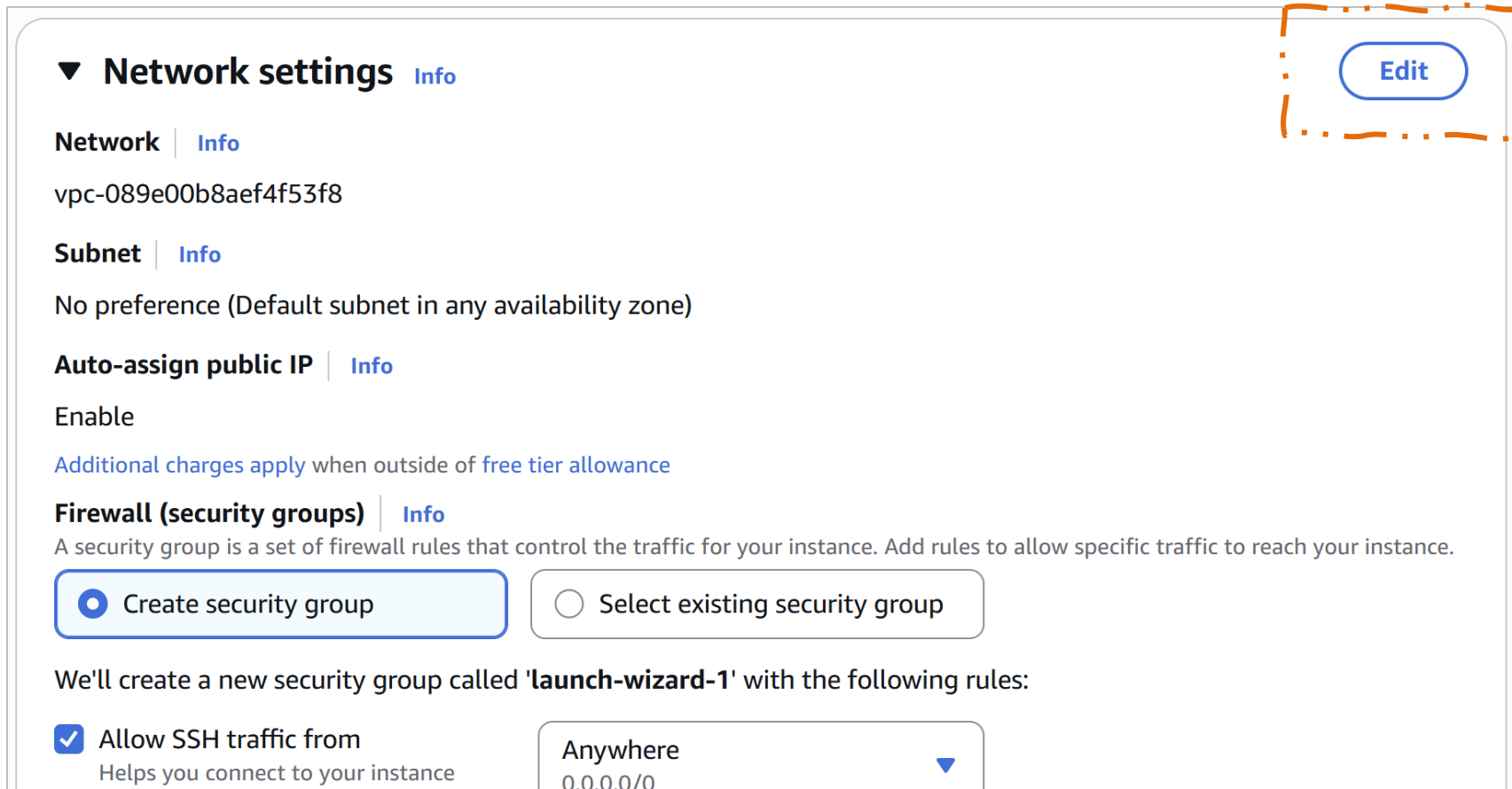
vockey



[Create new key pair](#)

Create an Instance – Network settings

- Network settings → Edits



▼ **Network settings** [Info](#)

Network | [Info](#)
vpc-089e00b8aef4f53f8

Subnet | [Info](#)
No preference (Default subnet in any availability zone)

Auto-assign public IP | [Info](#)
Enable
[Additional charges apply](#) when outside of [free tier allowance](#)

Firewall (security groups) | [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called '**launch-wizard-1**' with the following rules:

☒ Allow SSH traffic from
Helps you connect to your instance

Anywhere
0.0.0.0/0 ▼

Create an Instance – Network settings

- Network settings
 - VPC
 - Subnet
 - Auto-assign public IP

▼ Network settings [Info](#)

VPC - required | [Info](#)

vpc-089e00b8aef4f53f8 (default) ▼
172.31.0.0/16

Subnet | [Info](#)

subnet-031d587dea9244a5a ▼
VPC: vpc-089e00b8aef4f53f8 Owner: 211125458985
Availability Zone: us-east-1a (use1-az2) Zone type: Availability Zone
IP addresses available: 4091 CIDR: 172.31.80.0/20

Auto-assign public IP | [Info](#)

Enable ▼

[Additional charges apply](#) when outside of [free tier allowance](#)

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

Create an Instance – Configure storage

- Disk at 8 GiB

▼ **Configure storage** [Info](#)

Advanced

1x GiB ▼ Root volume, 3000 IOPS, Not encrypted

Add new volume

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

🕒 Click refresh to view backup information

↻

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

Create an Instance – Launch instance

- Summary panel of the right side
- Review Instance information
- Click **Launch instance**

▼ Summary

Number of instances | [Info](#)

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-0360c520857e3138f

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#)

Review our machine

- EC2 → Instances

The screenshot displays the AWS Management Console interface for the EC2 service. On the left, the navigation pane shows 'EC2' as the selected category, with sub-links for 'Dashboard', 'AWS Global View', 'Events', 'Instances', 'Instance Types', and 'Launch Templates'. The 'Instances' link is highlighted. The main content area is titled 'Instances (1)' and includes a 'Last updated' timestamp, a 'Connect' button, and buttons for 'Instance state', 'Actions', and 'Launch instances'. Below these are filter controls, including a 'Saved filter sets' dropdown and a search bar. A filter is applied: 'Instance state = running'. The instance list table below shows one instance:

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability
☐	joe11111	i-04c3472282fbe36ff	Running	t3.micro	Initializing	us-east-1a

- Select our instance

Review our machine

- Instance summary panel

Instance summary for **i-04c3472282fbe36ff (joe111111)** Info

Connect

Instance state

Actions

Updated less than a minute ago

Instance ID
 i-04c3472282fbe36ff

IPv6 address
—

Hostname type
IP name: ip-172-31-94-183.ec2.internal

Answer private resource DNS name
—

Auto-assigned IP address
 52.55.65.51 [Public IP]

IAM role
—

Public IPv4 address
 52.55.65.51 | [open address](#)

Instance state
 Running

Private IP DNS name (IPv4 only)
 ip-172-31-94-183.ec2.internal

Instance type
t3.micro

VPC ID
 vpc-089e00b8aef4f53f8

Subnet ID
 subnet-031d587dea9244a5a

Private IPv4 addresses
 172.31.94.183

Public DNS

ec2-52-55-65-51.compute-1.amazonaws.com | [open address](#)

Elastic IP addresses
—

AWS Compute Optimizer finding

Auto Scaling Group name
—



Manage EC2 Security

- Select Security → Select Security groups

Details	Status and alarms	Monitoring	Security	Networking	Storage	Tags
▼ Security details IAM role – Security groups sg-084935b0cf5d54b06 (launch-wizard-8) Owner ID 211125458985 Launch time Tue Sep 01 2026 10:09:45 GMT+0700 (Indochina Time)						

Manage EC2 Security

- Actions → Edit inbound rules

sg-084935b0cf5d54b06 - launch-wizard-8 Actions ▲

Details

Security group name launch-wizard-8	Security group ID sg-084935b0cf5d54b06	Description launch-wizard-8 created 2026-09-01T03:09:10.496Z
Owner 211125458985	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry

- Edit inbound rules
- Edit outbound rules
- Share security group
- Manage tags
- Copy to new security group
- Delete security groups

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

Inbound rules (1)

 Manage tags Edit inbound rules

Manage EC2 Security

- Click **Add rule** → **Add Rule** → **Save rules**

Inbound rules [Info](#)

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info
sgr-0ede953b692a7650f	SSH ▼	TCP	22	Cu... ▼
—	All ICMP - IPv4 ▼	ICMP	All	An... ▼
—	HTTP ▼	TCP	80	An... ▼

Add rule

0.0.0.0/0 ✕

0.0.0.0/0 ✕

0.0.0.0/0 ✕

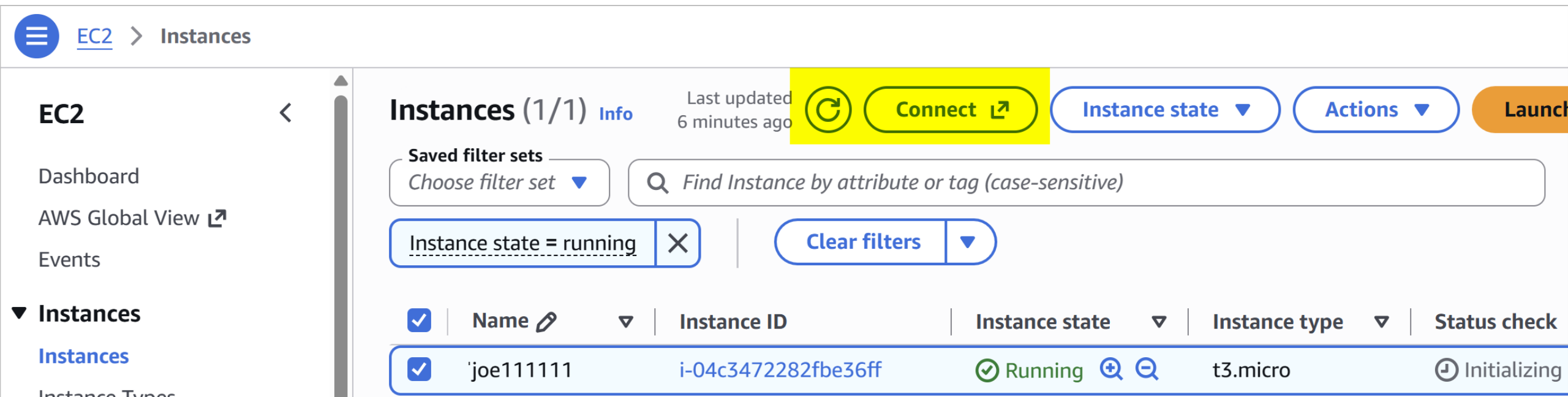
Manage EC2 Security

Inbound rules (3)								Manage tags	Edit inbound rules
Search							< 1 > 		
☐	Name ▼	Security group rule ID ▼	IP version ▼	Type ▼	Protocol ▼				
☐	-	sgr-0bd25dd9432c2f9ae	IPv4	HTTP	TCP				
☐	-	sgr-0ede953b692a7650f	IPv4	SSH	TCP				
☐	-	sgr-09959f518b115bf9a	IPv4	All ICMP - IPv4	ICMP				

- SSH from 0.0.0.0/0
- ICMP from 0.0.0.0/0
- HTTP from 0.0.0.0/0

Access your Machine

- EC2 → Instances → Select your Instance
- Click Connect



The screenshot shows the AWS Management Console interface for the EC2 Instances page. The left sidebar contains navigation links for EC2, Dashboard, AWS Global View, Events, and Instances. The main content area displays the 'Instances (1/1)' section with a 'Connect' button highlighted in yellow. Below the 'Connect' button, there are filter sets and a table of instances.

Instances (1/1) Info Last updated 6 minutes ago Refresh Connect Instance state Actions Launch

Saved filter sets
Choose filter set ▼

Find Instance by attribute or tag (case-sensitive)

Instance state = running X Clear filters ▼

☒	Name ✎ ▼	Instance ID	Instance state ▼	Instance type ▼	Status check
☒	joe111111	i-04c3472282fbe36ff	Running 🔍 🔍	t3.micro	Initializing 🕒

Access your Machine

```
Expanded Security Maintenance for Applications is not enabled.
```

```
0 updates can be applied immediately.
```

```
Enable ESM Apps to receive additional future security updates.
```

```
See https://ubuntu.com/esm or run: sudo pro status
```

```
The list of available updates is more than a week old.
```

```
To check for new updates run: sudo apt update
```

```
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.
```

```
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.
```

```
To run a command as administrator (user "root"), use "sudo <command>".
```

```
See "man sudo_root" for details.
```

```
ubuntu@ip-172-31-94-183:~$
```

i-04c3472282fbe36ff (joe111111)

PublicIPs: 52.55.65.51 PrivateIPs: 172.31.94.183

Our console is
ready

Add Web Server

- Update Linux Machine

```
$ sudo apt update && sudo apt upgrade -y
```

- Install Nginx Web Server

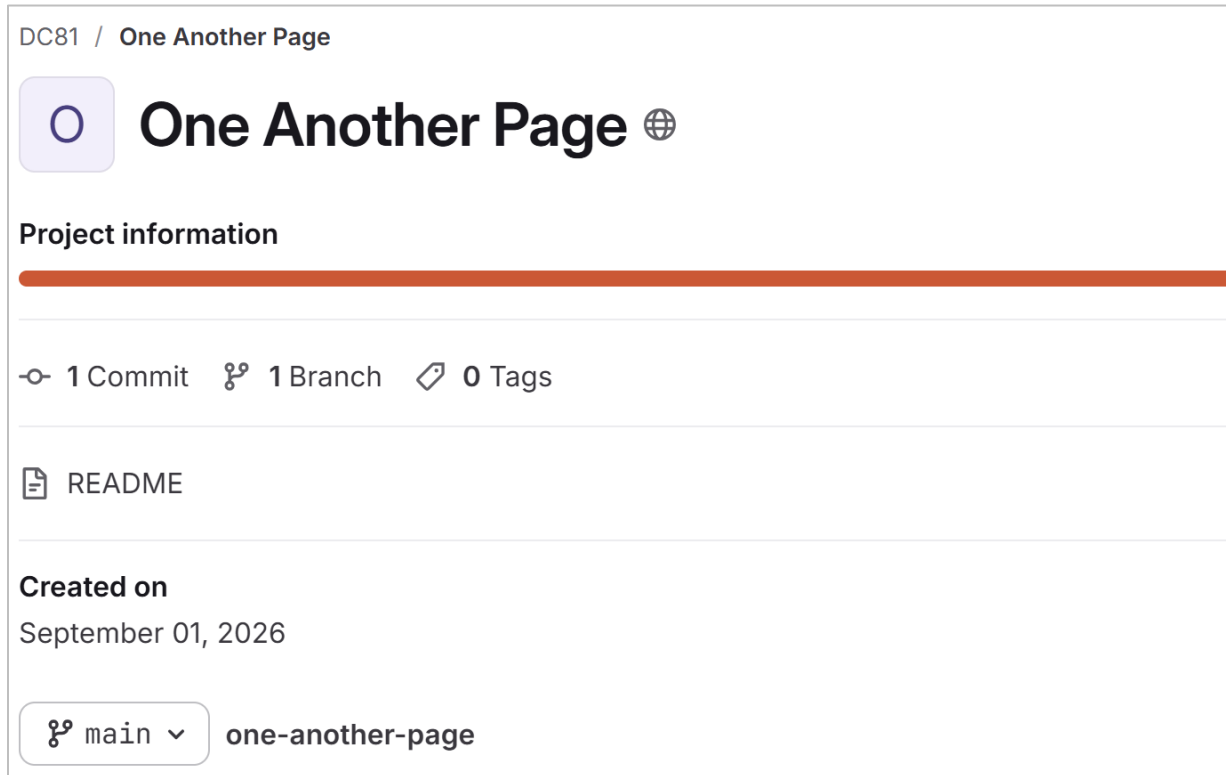
```
$ sudo apt install nginx -y
```

- Review Status

```
$ sudo systemctl status nginx
```

Add Web Content

- We will use One Another Page
<https://gitlab.com/dc81/one-another-page>



Add Web Content

- Install One Another Page

```
$ curl -sL https://gitlab.com/dc81/one-another-page/-/raw/main/install.sh | sudo bash
```

```
ubuntu@ip-172-31-94-183:~$  
ubuntu@ip-172-31-94-183:~$ curl -sL https://gitlab.com/dc81/one-another-page/-/raw/main/install.sh | sudo bash  
Downloading One Another page and configuring Nginx...  
✅ Configuration updated successfully! You can now check your web browser.  
ubuntu@ip-172-31-94-183:~$
```

- Restart Web Server

```
$ sudo systemctl restart nginx
```

Access Web Server

- EC2 → Instances → Instance Summary
- Public IPv4 address → Click Copy
- Open Browser and type URL →

`http://<IP>:80`



Ready to Launch

Ubuntu 24.04 Web Server is perfectly running on AWS. The setup is complete and ready for your next big project.

CS446/IT332

Document Root (Path):
`/var/www/html`

Server IP Address:
`172.31.94.183`

An aerial photograph of a lush green landscape featuring terraced rice fields. A light-colored, winding river or path cuts through the terraces, which are arranged in a series of concentric, undulating patterns. The fields are filled with water, reflecting the sky and surrounding vegetation. The overall scene is a harmonious blend of natural topography and human agricultural engineering.

Manage EC2 Elasticity

Stop the instance

- EC2 → Instance → Instance summary
- Instance state → Stop instance

Instance summary for i-04c3472282fbe36ff (joe111111) [Info](#)

 [Connect ↗](#) [Instance state ▲](#) [Actions ▼](#)

Updated less than a minute ago

Instance ID
 i-04c3472282fbe36ff

IPv6 address
—

Private IP address
172.31.1.1 | [open address ↗](#)

Public IP address
 ec2-52-1.amazonaws.com

Hostname type **Private IP DNS name (IPv4 only)**

Actions menu:

- Stop instance
- Start instance
- Reboot instance
- Hibernate instance
- Terminate (delete) instance

Stop the instance

- Click **Stop** to Confirm

Stop instance

Stopping your instance allows you to reduce costs, modify settings, and troubleshoot problems.

Instance ID ↗	Stop protection	Result 🚩
i-04c3472282fbe36ff (joe111111)	Disabled	✔ Can stop

▶ **Associated resources**
You will continue to incur charges for these resources while the instance is stopped

⚠ **You will be billed for associated resources**

After you stop the instance, you are no longer charged usage or data transfer fees for it. However, you will still be billed for associated Elastic IP addresses and EBS volumes.

Skip OS shutdown
This option skips the graceful OS shutdown process. Use only when your instance must be stopped immediately, such as during an emergency or failover.

☐ Skip OS shutdown

[Cancel](#)[Stop](#)

Resize the instance

- Wait until our instance is stopped
- Select our instance
- Action → Instance settings → Change instance type

The screenshot displays the AWS Management Console interface for an EC2 instance. At the top, a green notification bar states: "Successfully initiated stopping of i-04c3472282fbe36ff". Below this, the "Instance summary for i-04c3472282fbe36ff (joe111111)" is shown, including a refresh icon, a "Connect" button, and a dropdown menu for "Instance state". To the right, the "Actions" button is highlighted in yellow. A dropdown menu is open from "Actions", listing "Instance diagnostics", "Instance settings" (highlighted in yellow), "Networking", "Security", "Image and templates", and "Storage". On the left side of the console, under the "Instance configuration" section, the "Change instance type" button is also highlighted in yellow, with a mouse cursor hovering over it.

Resize the instance

- New instance type → t3.small → Change instance type

☰

EC2 > Instances > i-04c3472282fbe36ff > Change instance type

Current instance type

t3.micro

New instance type

🔍 t3.small

✕

Only instance types with x86 (32-bit) / (64-bit) architecture can be chosen.

☒ EBS-optimized

EBS-optimized is enabled by default for this instance type

▼ Instance type comparison

Attribute	t3.micro	t3.small
On-Demand Linux pricing	0.0104 USD per Hour	0.0208 USD per Hour
On-Demand Windows pricing	0.0196 USD per Hour	0.0392 USD per Hour
vCPUs	2 (1 core)	2 (1 core)
Memory (MiB)	1024	2048



Resize the instance

- Click **Change instance type**
- Instance changed

✓ Instance type changed successfully

Instance summary for i-04c3472282fbe36ff (joe111111) [Info](#)



Connect 

Instance state ▼

Actions ▼

Updated less than a minute ago

Instance ID  i-04c3472282fbe36ff	Public IPv4 address —	Private IPv4 addresses  172.31.94.183
IPv6 address —	Instance state  Stopped	Public DNS —
Hostname type IP name: ip-172-31-94-183.ec2.internal	Private IP DNS name (IPv4 only)  ip-172-31-94-183.ec2.internal	Elastic IP addresses —
Answer private resource DNS name —	Instance type t3.small	



Resize the Storage

- EC2 → Instance
- Select our [Instance ID](#) → Scroll Down → [Storage](#) → Select [Volume ID](#)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Root device details

Root device name
 /dev/sda1

Root device type
EBS

EBS optimization
enabled

▼ Block devices

 Filter block devices

< 1 > 

☒	Volume ID	Device name	Volume size (GiB)	Volume State	Attachment s
☒	vol-0f6e7f9dea9308944	/dev/sda1	8	 In-use	 Attached

Resize the Storage

- Select our volume
- Actions → Modify volume

The screenshot shows the AWS Management Console interface for the 'Volumes' section. At the top, it says 'Volumes (1/1)' with an 'Info' link. To the right, it indicates 'Last updated less than a minute ago' with a refresh icon. There are buttons for 'Recycle Bin' and 'Actions'. Below these, there's a 'Saved filter sets' section with a 'Choose filter set' dropdown and a 'Search' input. A filter is applied: 'Volume ID = vol-0f6e7f9dea9308944'. A 'Clear filters' button is also present. The main table lists the volume, with the first row highlighted in yellow. The row contains a checkmark, the volume name, the ID 'vol-0f6e7f9dea9308944', the type 'gp3', and the size '8 G'. The 'Actions' dropdown menu is open, showing options: 'Modify volume' (highlighted), 'Create snapshot', 'Create snapshot lifecycle policy', 'Delete volume', 'Attach volume', 'Detach volume', and 'Force detach volume'.

Volumes (1/1) Info

Last updated less than a minute ago

Recycle Bin Actions

Saved filter sets
Choose filter set

Search

Volume ID = vol-0f6e7f9dea9308944 X Clear filters

☒	Name	Volume ID	Type	Size
☒		vol-0f6e7f9dea9308944	gp3	8 G

Modify volume
Create snapshot
Create snapshot lifecycle policy
Delete volume
Attach volume
Detach volume
Force detach volume

Resize the Storage

- Change from 8 to 20
- Click **Modify** → Confirm **Modify**

Modify vol-0f6e7f9dea9308944?

✕

If you are increasing the size of the volume, you must extend the file system to the new size of the volume. You can only do this when the volume enters the optimizing state. For more information see [Extend the file system after resizing an EBS volume](#).

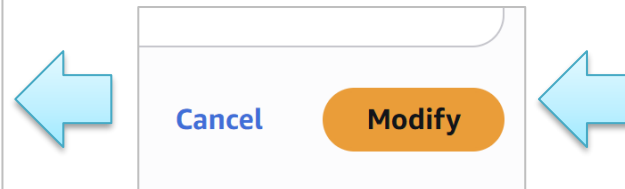
The modification might take a few minutes to complete.

You are charged for the new volume configuration after volume modification starts. For pricing information, see [Amazon EBS Pricing](#).

Are you sure that you want to modify vol-0f6e7f9dea9308944?

Cancel

Modify



Modify volume

Info

Modify the type, size, and performance of an EBS volume.

Volume details

Volume ID

vol-0f6e7f9dea9308944

Volume type

Info

General Purpose SSD (gp3)

Size (GiB)

Info

20

Min: 1 GiB, Max: 65536 GiB.

IOPS

Info

3000

Min: 3000 IOPS, Max: 80000 IOPS.

Throughput (MiB/s)

Info

125

Min: 125 MiB, Max: 2000 MiB. Baseline: 125 MiB/s.

Resize the Storage

- Click refresh until it expands to 20G

Requested volume modification for volume `vol-0f6e7f9dea9308944`.
The volume is being modified.

Volumes (1) [Info](#)

Last updated 1 minute ago 

[Recycle Bin](#) [Actions ▼](#) [Create v](#)

Saved filter sets
[Choose filter set ▼](#)

[X](#) [Clear filters ▼](#)

☐	Name 	Volume ID	Type	Size	IOPS	Throughput
☐		vol-0f6e7f9dea9308944	gp3	20 GiB	3000	125

Start our Instance

- EC2 → Instance
- Select our Instance
- Instance state → Start instance

The screenshot displays the AWS Management Console's EC2 Instances page. At the top, it shows 'Instances (1/1)' with an 'Info' link. Below this, there's a 'Last updated' timestamp and a 'Connect' button. A 'Saved filter sets' section includes a dropdown for 'Choose filter set' and a search bar for 'Find Instance'. A filter is applied: 'Instance state = running'. The 'Instance state' dropdown menu is open, showing options: 'Stop instance', 'Start instance' (highlighted with a yellow box and a hand cursor), 'Reboot instance', 'Hibernate instance', and 'Terminate (delete) instance'. The 'Actions' dropdown is also visible. A 'Launch instances' button is on the right. The instance list table has columns for 'Name', 'Instance ID', 'Instance state', and 'Instance type'. One instance is listed: 'joe111111' with ID 'i-04c3472282fbe36ff', state 'Stopped' (highlighted with a yellow box), and type 't3.small' (highlighted with a yellow box).

Name	Instance ID	Instance state	Instance type
joe111111	i-04c3472282fbe36ff	Stopped	t3.small

Access to our Machine

- Instance summary → Connect

Instance summary for i-04c3472282fbe36ff (joe111111) [Info](#)

 [Connect](#) [Instance state](#) [Actions](#)

Updated 7 minutes ago

Instance ID  i-04c3472282fbe36ff	Public IPv4 address  52.87.121.21 open address
IPv6 address —	Instance state  Running
Hostname type IP name: ip-172-31-94-183.ec2.internal	Private IP DNS name (IPv4 only)  ip-172-31-94-183.ec2.internal
Answer private resource DNS name —	Instance type t3.small

Review your Machine Changed

- Review disk and memory changed

```
$ df -h
```

```
$ free -h
```

```
ubuntu@ip-172-31-94-183:~$  
ubuntu@ip-172-31-94-183:~$ df -h  
Filesystem      Size  Used Avail Use% Mounted on  
/dev/root        19G   2.1G   17G   12% /  
tmpfs            954M    0   954M    0% /dev/shm  
tmpfs            382M   872K   381M    1% /run  
tmpfs            5.0M    0    5.0M    0% /run/lock  
efivarfs         128K    3.3K   120K    3% /sys/firmware/efi/efivars  
/dev/nvme0n1p16  881M    94M   726M   12% /boot  
/dev/nvme0n1p15  105M    6.2M    99M    6% /boot/efi  
tmpfs            191M    12K   191M    1% /run/user/1000  
ubuntu@ip-172-31-94-183:~$ free -h  
              total        used        free      shared  buff/cache   available  
Mem:          1.9Gi         365Mi       1.4Gi         2.7Mi        256Mi       1.5Gi  
Swap:          0B           0B           0B  
ubuntu@ip-172-31-94-183:~$
```

An aerial photograph of a lush green landscape featuring terraced rice fields. A light-colored, winding river or path cuts through the terraces, which are arranged in a series of concentric, undulating patterns. The fields are filled with water, reflecting the sky and surrounding vegetation. The overall scene is a harmonious blend of natural topography and human agricultural engineering.

Create and Attach New EBS Volume

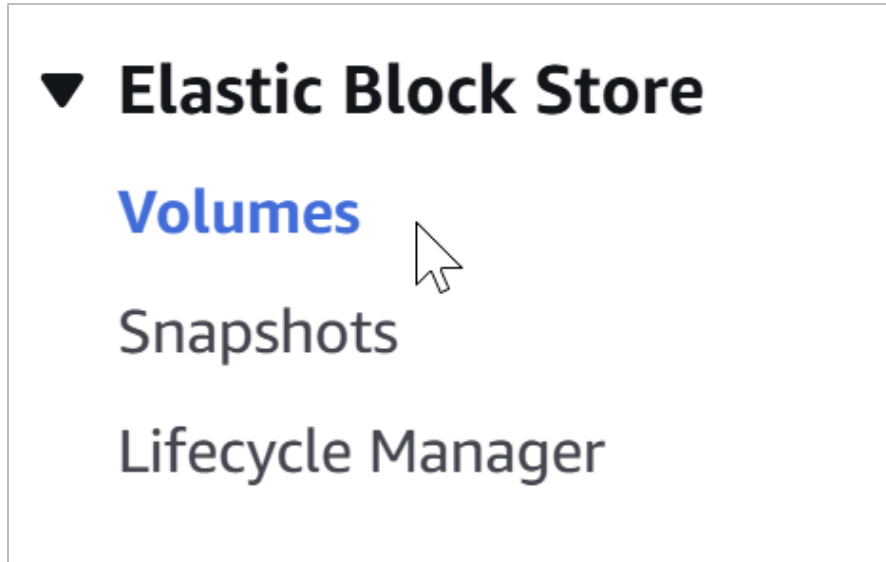
Review Availability Zone

- EC2 → Select our instance
- Network → Review Availability Zone

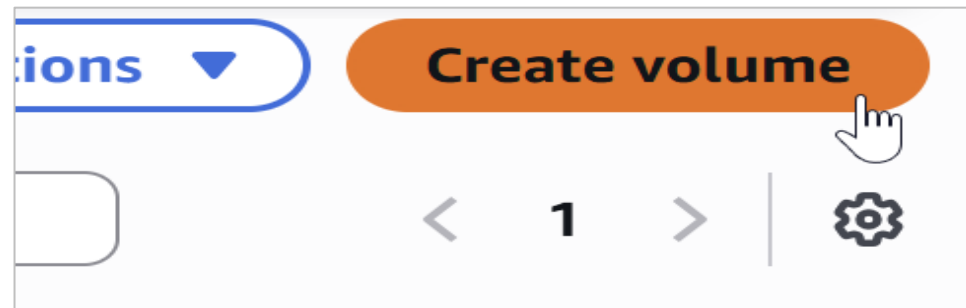
Details		Status and alarms		Monitoring		Security		Networking		Storage		Tags	
VPC ID  vpc-089e00b8aef4f53f8 				Subnet ID  subnet-031d587dea9244a5a 				Availability zone  us-east-1a					
Availability zone ID  use1-az2				Outpost ID –									
▼ IP addresses Info													
Public IPv4 address  52.87.121.21 open address 				Private IPv4 addresses  172.31.94.183				IPv6 addresses –					
Secondary private IPv4 addresses –				Carrier IP addresses (ephemeral) –									

Create EBS Volume

- Elastic Block Store → Volumes



- Click Create volume



Create EBS Volume - Size

- Size = 1

Volume settings

Volume type | [Info](#)

General Purpose SSD (gp3)

Size (GiB) | [Info](#)

1

Min: 1 GiB, Max: 16384 GiB.

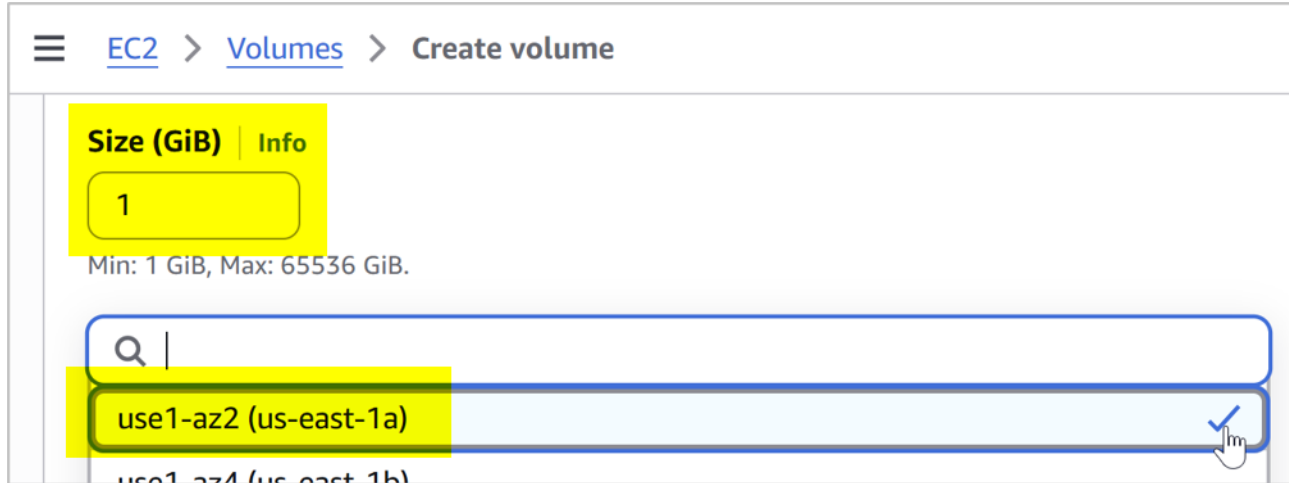
IOPS | [Info](#)

3000

Min: 3000 IOPS, Max: 16000 IOPS.

Create EBS Volume – Availability Zone

- Choose Availability Zone



EC2 > Volumes > Create volume

Size (GiB) | Info

1

Min: 1 GiB, Max: 65536 GiB.

use1-az2 (us-east-1a)

- Must with the same with EC2 Availability Zone

Details			Status and alarms			Monitoring			Security			Networking			Storage		
VPC ID vpc-089e00b8aef4f53f8			Subnet ID subnet-031d587dea9244a5a			Availability zone us-east-1a			Availability zone ID use1-az2			Outpost ID –					

Create EBS Volume – Tag

- Add Tag as name for EBS identification
- Key: Name
- Value:

Tags - *optional* [Info](#)

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value.

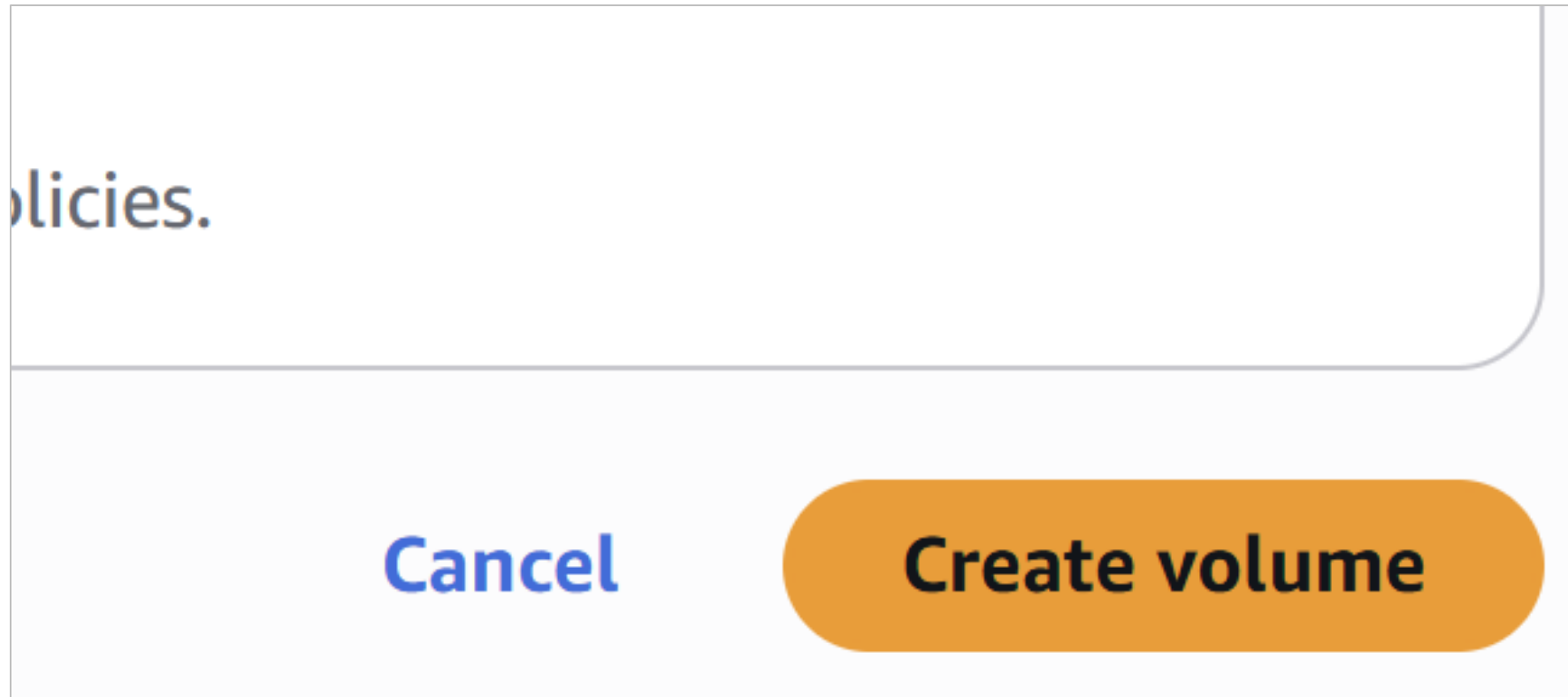
Key

Value - *optional*

Create EBS Volume

- Create volume



Review Volume Created

- Clear Filters
- Refresh
- Review your Volumes

Volumes (1/1) [Info](#) Last updated 1 minute ago  [↗ Recycle Bin](#)

Saved filter sets
Choose filter set ▼

vol-0f6e7f9dea9308944 ✕ Clear filters ▼



Volumes (2) [Info](#) Last updated 1 minute ago  [↗ Recycle Bin](#) [Actions ▼](#)

Saved filter sets
Choose filter set ▼

☐	Name 	Volume ID	Type	Size	IOPS	Throughput
☐	joe111111-disk	vol-083971abfca444b35	gp3	1 GiB	3000	125
☐		vol-0f6e7f9dea9308944	gp3	20 GiB	3000	125

Attach Volume to our Instance

- Select newly created volume
- Action → Attach volume

Volumes (1/2) [Info](#) Last updated 5 minutes ago  [Recycle Bin](#) [Actions ▲](#)

Saved filter sets [Choose filter set ▼](#)

	Name 	Volume ID 	Type 	Size
☒	joe111111-disk	vol-083971abfca444b35	gp3	1 G
☐		vol-0f6e7f9dea9308944	gp3	20



Modify volume

Create snapshot

Create snapshot lifecycle policy

Delete volume

Attach volume

Detach volume

Attach Volume to our Instance

- Select running instance → /dev/sdf → Attach volume

Availability Zone
use1-az2 (us-east-1a)

Instance | Info
i-04c3472282fbe36ff
(joe111111) (running)
Only instances in the same Availability Zone as the selected volume are displayed.

Device name | Info
/dev/sdf
Recommended device names for Linux: /dev/sda1 for root volume. /dev/sd[f-p] for data volumes.

i Newer Linux kernels may rename your devices to **/dev/xvdf** through **/dev/xvdp** internally, even when the device name entered here (and shown in the details) is **/dev/sdf** through **/dev/sdp**.

[Cancel](#) [Attach volume](#)

Attach Volume to our Instance

- EC2 → Instance → Select our instance
- Scroll down to Storage

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Root device details

Root device name
 /dev/sda1

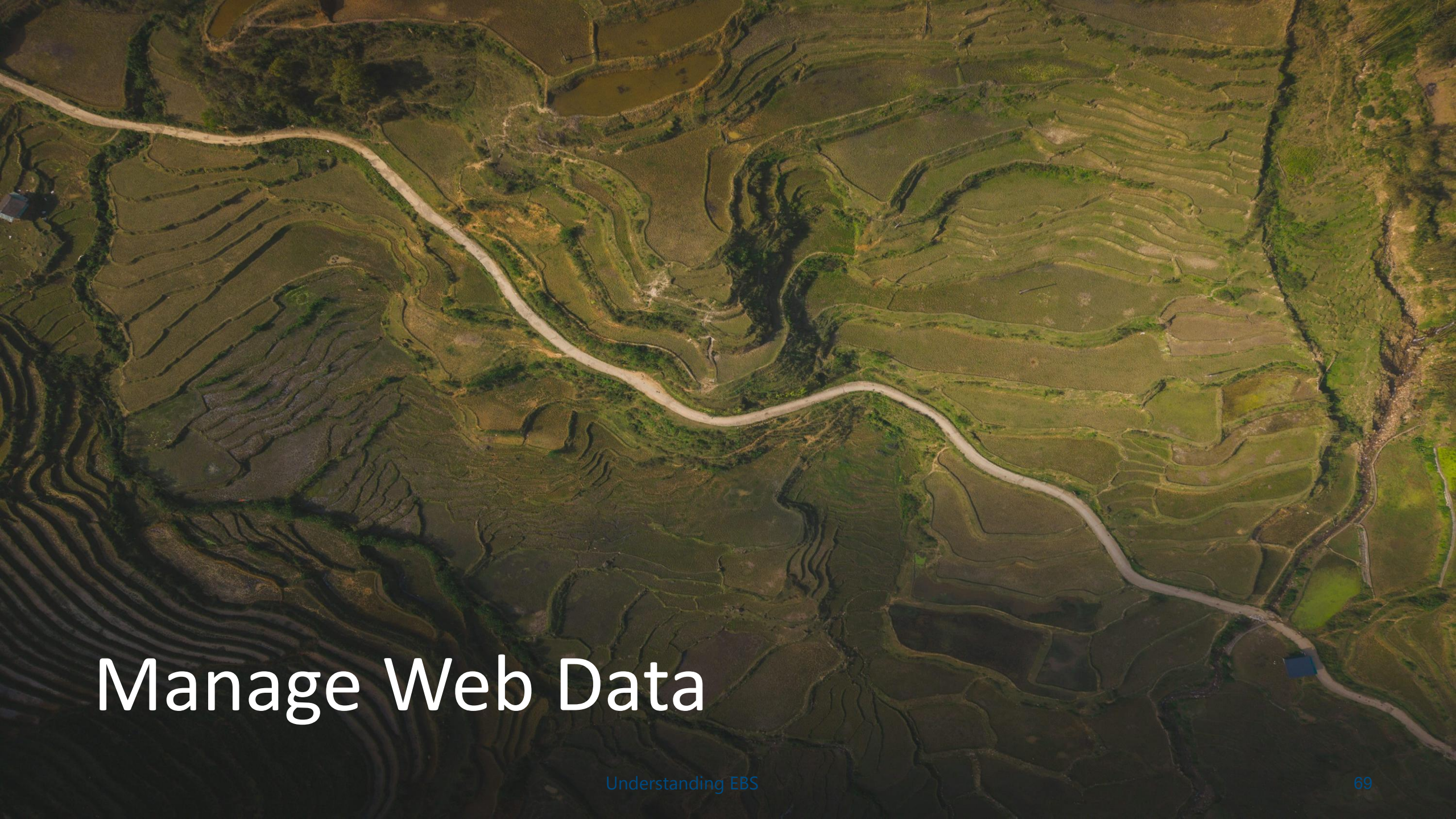
Root device type
EBS

EBS optimization
enabled

▼ Block devices

< 1 > 

☐	Volume ID	Device name	Volume size (GiB)	Volume State	Attachment s
☐	vol-0f6e7f9dea9308944	/dev/sda1	20	✓ In-use	✓ Attached
☐	vol-083971abfca444b35	/dev/sdf	1	✓ In-use	✓ Attached

An aerial photograph of a terraced rice field landscape. The terraces are carved into the hillsides, creating a series of concentric, wavy lines. A light-colored, winding road or path runs through the center of the image, following the contours of the terraces. The fields are filled with water, reflecting the sky and creating a shimmering effect. The overall color palette is dominated by greens and browns, with the white text providing a stark contrast.

Manage Web Data

Access to our Machine

- Instance summary → Connect

Instance summary for i-04c3472282fbe36ff (joe111111) [Info](#)

 [Connect](#) [Instance state](#) [Actions](#)

Updated 7 minutes ago

Instance ID  i-04c3472282fbe36ff	Public IPv4 address  52.87.121.21 open address
IPv6 address —	Instance state  Running
Hostname type IP name: ip-172-31-94-183.ec2.internal	Private IP DNS name (IPv4 only)  ip-172-31-94-183.ec2.internal
Answer private resource DNS name —	Instance type t3.small

Review New Disk Attached

- Inspect newly disk attached → Looking for 1G

\$ lsblk

```
ubuntu@ip-172-31-94-183:~$  
ubuntu@ip-172-31-94-183:~$ lsblk  
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS  
loop0                 7:0    0 28.2M  1 loop /snap/amazon-ssm-agent/13009  
loop1                 7:1    0   74M  1 loop /snap/core22/2411  
loop2                 7:2    0 49.3M  1 loop /snap/snapd/26865  
nvme0n1              259:0    0   20G  0 disk  
├─nvme0n1p1          259:1    0   19G  0 part /  
├─nvme0n1p14         259:2    0    4M  0 part  
├─nvme0n1p15         259:3    0 106M  0 part /boot/efi  
└─nvme0n1p16         259:4    0 913M  0 part /boot  
nvme1n1              259:5    0    1G  0 disk  
ubuntu@ip-172-31-94-183:~$
```



Format and Mount Disk

- Format Disk

```
$ sudo mkfs -t ext4 /dev/nvme1n1
```

- Make Web Data Directory

```
$ sudo mkdir /data
```

- Mount Disk to Web Data

```
$ sudo mount /dev/nvme1n1 /data
```

```
ubuntu@ip-172-31-94-183:~$ sudo mkfs -t ext4 /dev/nvme1n1
mke2fs 1.47.0 (5-Feb-2023)
Creating filesystem with 262144 4k blocks and 65536 inodes
Filesystem UUID: 08de80e4-5b41-4fb6-a2f5-83cc6822196f
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

ubuntu@ip-172-31-94-183:~$ sudo mkdir /data
ubuntu@ip-172-31-94-183:~$ sudo mount /dev/nvme1n1 /data
```



Format and Mount Disk

- Review mounted disk

\$ lsblk

- Review disk capacity

\$ df -h

```
ubuntu@ip-172-31-94-183:~$ lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0                 7:0    0  28.2M  1 loop /snap/amazon-ssm-agent/13009
loop1                 7:1    0   74M  1 loop /snap/core22/2411
loop2                 7:2    0  49.3M  1 loop /snap/snapd/26865
nvme0n1              259:0    0   20G  0 disk
├─nvme0n1p1          259:1    0   19G  0 part /
├─nvme0n1p14         259:2    0    4M  0 part
├─nvme0n1p15         259:3    0  106M  0 part /boot/efi
└─nvme0n1p16         259:4    0   913M  0 part /boot
nvme1n1              259:5    0    1G  0 disk /data

ubuntu@ip-172-31-94-183:~$ df -h
Filesystem            Size  Used Avail Use% Mounted on
/dev/root              19G   2.1G   17G  12% /
tmpfs                  954M     0  954M   0% /dev/shm
tmpfs                  382M  880K  381M   1% /run
tmpfs                   5.0M     0   5.0M   0% /run/lock
efivarfs               128K   3.3K  120K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p16        881M   94M  726M  12% /boot
/dev/nvme0n1p15       105M   6.2M   99M   6% /boot/efi
tmpfs                  191M   12K  191M   1% /run/user/1000
/dev/nvme1n1           974M   24K  907M   1% /data
```

Relocate Web Data

- Change Web Server Configuration

```
$ sudo sed -i 's#/var/www/html#/data#g' /etc/nginx/sites-enabled/default
```

- Check your edit

```
$ grep data /etc/nginx/sites-enabled/default
```

```
ubuntu@ip-172-31-94-183:~$ sudo sed -i 's#/var/www/html#/data#g' /etc/nginx/sites-enabled/default
ubuntu@ip-172-31-94-183:~$ grep data /etc/nginx/sites-enabled/default
root /data;
```

- Review your Web Server Configuration

```
$ sudo nginx -t
```

```
ubuntu@ip-172-31-94-183:~$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
ubuntu@ip-172-31-94-183:~$
```

- If your Web Server has any error message – Fix it first

Relocate Web Data

- Restart Web Server

```
$ sudo cp /var/www/html/index.html /data/index.html
```

```
$ ls -l /data/index.html
```

```
ubuntu@ip-172-31-94-183:~$  
ubuntu@ip-172-31-94-183:~$ sudo cp /var/www/html/index.html /data/index.html  
ubuntu@ip-172-31-94-183:~$ ls -l /data/index.html  
-rw-r--r-- 1 root root 2403 Sep  1 06:07 /data/index.html  
ubuntu@ip-172-31-94-183:~$
```

Relocate Web Data

- Restart Web Server

```
$ sudo systemctl restart nginx
```

- Review Web Service

```
$ sudo systemctl status nginx
```

```
ubuntu@ip-172-31-94-183:~$ sudo systemctl restart nginx
ubuntu@ip-172-31-94-183:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-09-01 06:05:24 UTC; 7s ago
     Docs: man:nginx(8)
  Process: 2299 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
  Process: 2302 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
 Main PID: 2303 (nginx)
    Tasks: 3 (limit: 2209)
  Memory: 2.5M (peak: 2.7M)
     CPU: 15ms
   CGroup: /system.slice/nginx.service
           └─2303 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─2304 "nginx: worker process"
               └─2305 "nginx: worker process"
```



Rewrite our web content

- Access our web server

`http://<IP>:80`



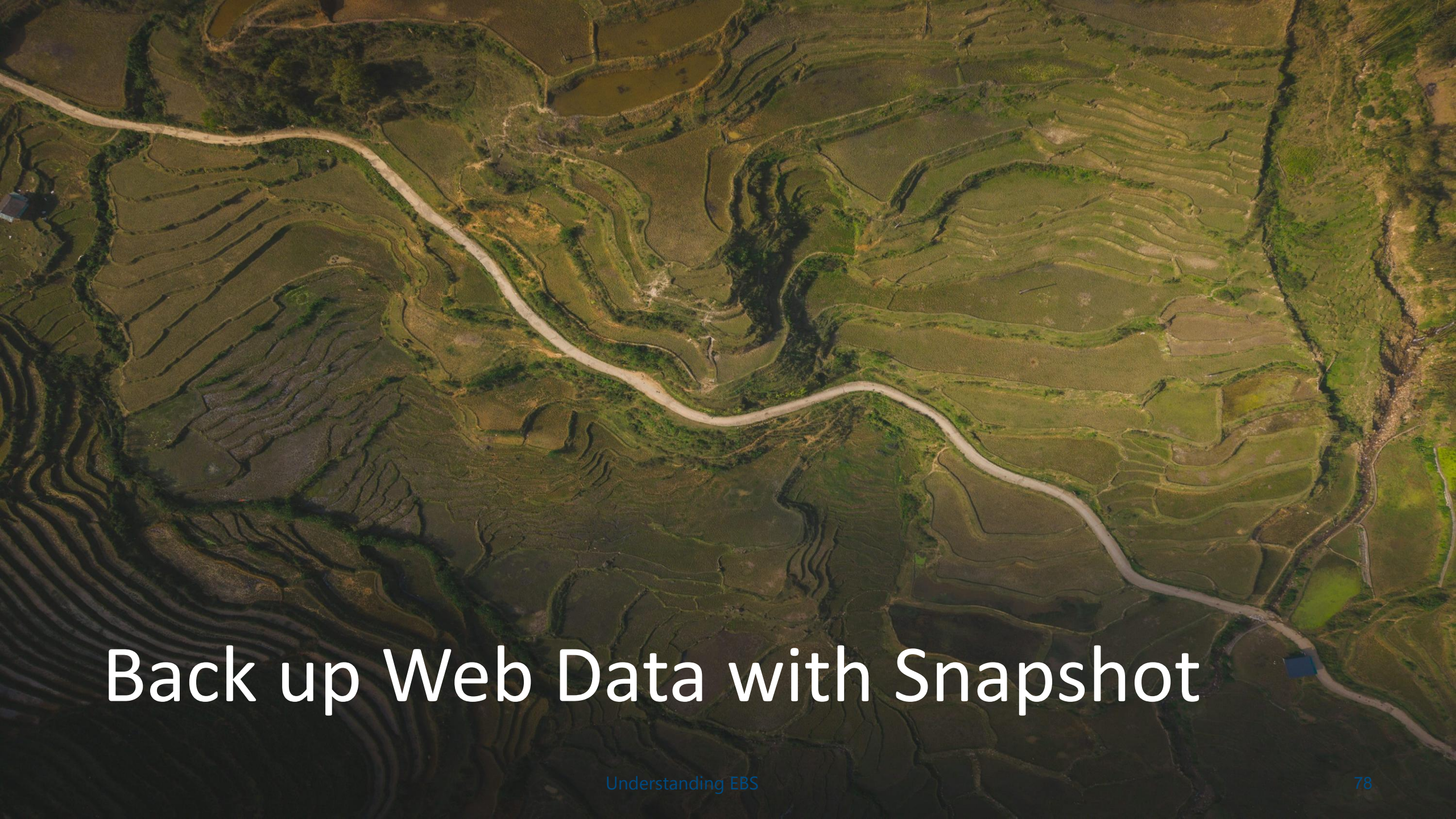
Ready to Launch

Ubuntu 24.04 Web Server is perfectly running on AWS. The setup is complete and ready for your next big project.

CS446/IT332

Document Root (Path):
/data

Server IP Address:
172.31.94.183

An aerial photograph of a lush green landscape featuring terraced rice fields. A light-colored, winding river or path cuts through the terraces, which are arranged in concentric, stepped patterns. The fields are filled with water, reflecting the sky and surrounding vegetation. The overall scene is a harmonious blend of natural topography and human agricultural engineering.

Back up Web Data with Snapshot

Backup Web Data with EBS Snapshot

- Our Instance → Storage
- Select our web data disk

Storage

▼ Root device details

Root device name:  /dev/sda1

Root device type: EBS

EBS optimization: enabled

▼ Block devices

Filter block devices

	Volume ID	Device name	Volume size (GiB)	Volume State	Attachment
☐	vol-0f6e7f9dea9308944	/dev/sda1	20	 In-use	 Attach
☒	vol-083971abfca444b35	/dev/sdf	1	 In-use	 Attach

Backup Web Data with EBS Snapshot

- Select our volume → Action → Create snapshot

The screenshot shows the AWS Management Console 'Volumes' page. At the top, it says 'Volumes (1/1)' with an 'Info' link. To the right, it indicates 'Last updated less than a minute ago' with a refresh icon. There are buttons for 'Recycle Bin' and 'Actions'. Below this, there's a 'Saved filter sets' section with a dropdown 'Choose filter set' and a search bar. A filter is applied: 'Volume ID = vol-083971abfca444b35'. A 'Clear filters' button is next to it. The main table lists volumes. The first volume is selected (checkbox checked) and has the name 'joe111111-disk', ID 'vol-083971abfca444b35', type 'gp3', and size '1 G'. The 'Actions' dropdown menu is open, showing options: 'Modify volume', 'Create snapshot' (highlighted with a hand icon), 'Create snapshot lifecycle policy', 'Delete volume', 'Attach volume', 'Detach volume', and 'Force detach volume'.

☒	Name	Volume ID	Type	Size
☒	joe111111-disk	vol-083971abfca444b35	gp3	1 G

Backup Web Data with EBS Snapshot

- Description: Backup web data
- Tags: Name = backup-web-data → Create snapshot

Snapshot details

Description
Add a description for your snapshot

Backup web data

255 characters maximum.

Encryption [Info](#)
Not encrypted

Tags [Info](#)

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key	Value - optional	
Name	backup-web-data	Remove

[Add new tag](#)

You can add up to 49 more tags.

[Cancel](#)[Create snapshot](#)



Review EBS Snapshot

- EC2 → Elastic Block Store → Snapshots

AMI Catalog

▼ Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

▼ Network & Security

Snapshots (1) Info

Last updated less than a minute ago

↺ ↻

↗ Recycle Bin

Actions ▼

Create

Snapshot scope

Owned by me ▼

Q Search

☐	Name ✎ ▼	Snapshot ID ▼	Full snapshot size ▼	Volume size ▼	Description
☐	backup-web-d...	snap-0650a84c09ad1a860	52 MiB	1 GiB	Backup web data

Review EBS Snapshot

- Click Snapshot ID

snap-0650a84c09ad1a860 (backup-web-data)

Last updated
less than a minute ago

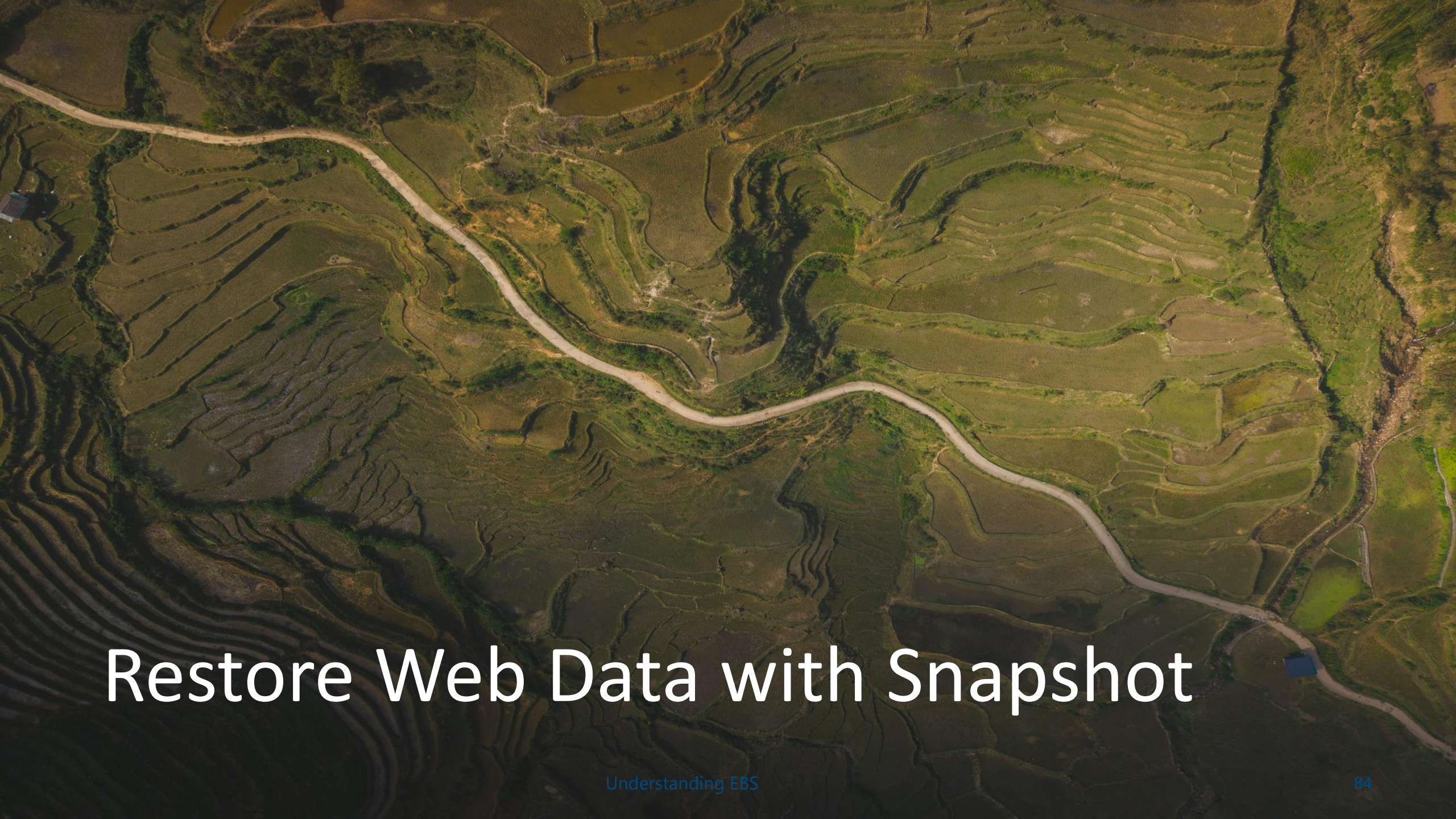
↺

Delete

Actions ▼

Details

Snapshot ID ☐ snap-0650a84c09ad1a860 (backup-web-data)	Full snapshot size ☐ 52 MiB	Progress ✔ 100%	Snapshot status ✔ Completed
Owner ☐ 211125458985	Started ☐ Tue Sep 01 2026 14:53:45 GMT+0700 (Indochina Time)	Product codes -	Fast snapshot restore -
Description ☐ Backup web data			
▼ Source volume			
Volume ID ☐ vol-083971abfca444b35	Volume size ☐ 1 GiB		

An aerial photograph of a lush, green landscape featuring terraced rice fields. A winding river or stream flows through the center of the image, reflecting the surrounding greenery. The terraces are arranged in a series of concentric, wavy lines, creating a complex, organic pattern. The lighting is bright, casting soft shadows that emphasize the contours of the fields.

Restore Web Data with Snapshot

Accidentally Remove our Web Content

- Remove our web content

```
$ sudo rm -rf /data/index.html
```

```
$ ls -l /data
```

BE CAREFUL OR YOU MAY ACCIDENTALLY WIPE YOUR SYSTEM

```
ubuntu@ip-172-31-94-183:~$  
ubuntu@ip-172-31-94-183:~$ sudo rm -rf /data/index.html  
ubuntu@ip-172-31-94-183:~$ ls -l /data  
total 0  
ubuntu@ip-172-31-94-183:~$
```

Accidentally Remove our Web Content

- Access your Web → Our web is no longer there

403 Forbidden

nginx/1.24.0 (Ubuntu)

Restore our Web Content from Snapshot

- EC2 → Elastic Block Store → Snapshots
- Select out snapshot
- Actions → Create volume from snapshot

The screenshot shows the AWS Management Console 'Snapshots' page. At the top, it says 'Snapshots (1/1)' with an 'Info' link. To the right, it indicates 'Last updated less than a minute ago' with a refresh icon. There are buttons for 'Recycle Bin' and 'Actions'. Below these, there's a 'Snapshot scope' dropdown set to 'Owned by me' and a search bar. A table lists snapshots with columns for selection, Name, Snapshot ID, and Full snapshot. One snapshot is highlighted in yellow: 'backup-web-d...' with ID 'snap-0650a84c09ad1a860' and size '52 MiB'. An 'Actions' menu is open for this snapshot, listing options: 'Create volume from snapshot' (highlighted with a hand cursor), 'Create image from snapshot', 'Copy snapshot', 'Launch copy duration calculator', and 'Delete snapshot'.

	Name	Snapshot ID	Full snapshot
☒	backup-web-d...	snap-0650a84c09ad1a860	52 MiB

- Create volume from snapshot
- Create image from snapshot
- Copy snapshot
- Launch copy duration calculator
- Delete snapshot

Restore our Web Content from Snapshot

- Check Disk Size

Volume settings

Snapshot ID
 snap-0650a84c09ad1a860 (backup-web-data)

Volume type | [Info](#)
General Purpose SSD (gp3) ▼

Size (GiB) | [Info](#)

Min: 1 GiB, Max: 65536 GiB.

Restore our Web Content from Snapshot

- Select Availability Zone → Same Location with our Instance

Availability Zone	Info
use1-az2 (us-east-1a)	

Restore our Web Content from Snapshot

- Provide tag with meaningful information

Tags - optional [Info](#)
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or to

Key	Value - optional	
Name	restore-web-data	Remove
Add new tag		

You can add up to 49 more tags.

- Click Create volume

[Cancel](#) [Create volume](#)

Review our restored snapshot

- Elastic Block Store → Volumes
- Clear any filters

Volumes (3) [Info](#)

Last updated less than a minute ago 

 **Recycle Bin**

Actions ▼

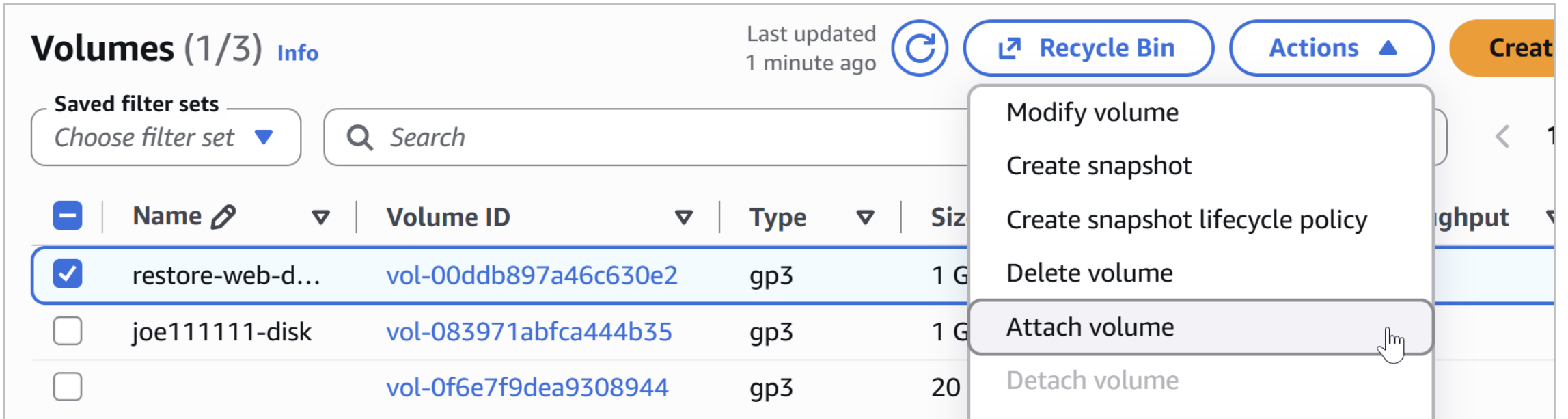
Saved filter sets
Choose filter set ▼

 Search

☐	Name  ▼	Volume ID ▼	Type ▼	Size ▼	IOPS ▼	Throughput ▼
☐	restore-web-d...	vol-00ddb897a46c630e2	gp3	1 GiB	3000	125
☐	joe111111-disk	vol-083971abfca444b35	gp3	1 GiB	3000	125
☐		vol-0f6e7f9dea9308944	gp3	20 GiB	3000	125

Attach Restored Volume

- Select volume that we restored from snapshot
- Actions → Attach volume



The screenshot shows the AWS Management Console 'Volumes' page. At the top, it says 'Volumes (1/3)' with an 'Info' link. To the right, it indicates 'Last updated 1 minute ago' with a refresh icon. There are buttons for 'Recycle Bin', 'Actions', and 'Create'. Below the header, there's a 'Saved filter sets' section with a dropdown 'Choose filter set' and a search bar. The main content is a table of volumes. The first volume, 'restore-web-d...', is selected with a checkbox. An 'Actions' menu is open for this volume, showing options: 'Modify volume', 'Create snapshot', 'Create snapshot lifecycle policy', 'Delete volume', 'Attach volume' (highlighted with a hand cursor), and 'Detach volume'.

	Name	Volume ID	Type	Size
☒	restore-web-d...	vol-00ddb897a46c630e2	gp3	1 G
☐	joe111111-disk	vol-083971abfca444b35	gp3	1 G
☐		vol-0f6e7f9dea9308944	gp3	20

Attach Restored Volume

- Instance →
Select our instance
- Device name →
`/dev/sdg`
- Attach Volume

[Cancel](#)[Attach volume](#)

EC2 > Volumes > vol-00ddb897a46c630e2 > Attach volume

Basic details

Volume ID
vol-00ddb897a46c630e2 (restore-web-data)

Availability Zone
use1-az2 (us-east-1a)

Instance | Info
i-04c3472282fbe36ff
(joe111111) (running)

Device name | Info
/dev/sdg

Review new volume in our instance

- EC2 → Instance → Storage (Refresh if needed)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Root device details

Root device name

 /dev/sda1

Root device type

EBS

EBS optimization

enabled

▼ Block devices

Q

Filter block devices

< 1

☐	Volume ID	Device name	Volume size (GiB)	Volume State
☐	vol-0f6e7f9dea9308944	/dev/sda1	20	 In-use
☐	vol-083971abfca444b35	/dev/sdf	1	 In-use
☐	vol-00ddb897a46c630e2	/dev/sdg	1	 In-use

Restore our Web Data

- List new disk attached

```
$ lsblk
```

- Make new web data directory

```
$ sudo mkdir /data2
```

- Mount restored disk to new web data

```
$ sudo mount /dev/nvme2n1 /data2
```

- Review our web data

```
$ ls -l /data2
```

Restore our Web Data

```
ubuntu@ip-172-31-94-183:~$ lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0                7:0      0 28.2M  1 loop /snap/amazon-ssm-agent/13009
loop1                7:1      0 28.4M  1 loop /snap/amazon-ssm-agent/13349
loop2                7:2      0   74M  1 loop /snap/core22/2411
loop3                7:3      0 49.3M  1 loop /snap/snapd/26865
nvme1n1              259:0     0    1G   0 disk
nvme0n1              259:1     0   20G   0 disk
├─nvme0n1p1          259:2     0   19G   0 part /
├─nvme0n1p14         259:3     0    4M   0 part
├─nvme0n1p15         259:4     0 106M   0 part /boot/efi
└─nvme0n1p16         259:5     0 913M   0 part /boot
nvme2n1              259:6     0    1G   0 disk
ubuntu@ip-172-31-94-183:~$ sudo mkdir /data2
ubuntu@ip-172-31-94-183:~$ sudo mount /dev/nvme2n1 /data2
ubuntu@ip-172-31-94-183:~$ ls -l /data2
total 20
-rw-r--r-- 1 root root 2403 Sep  1 06:07 index.html
drwx----- 2 root root 16384 Sep  1 05:49 lost+found
```

Restore our Web Data

- Restored our web data

```
$ sudo cp /data2/index.html /data/index.html
```

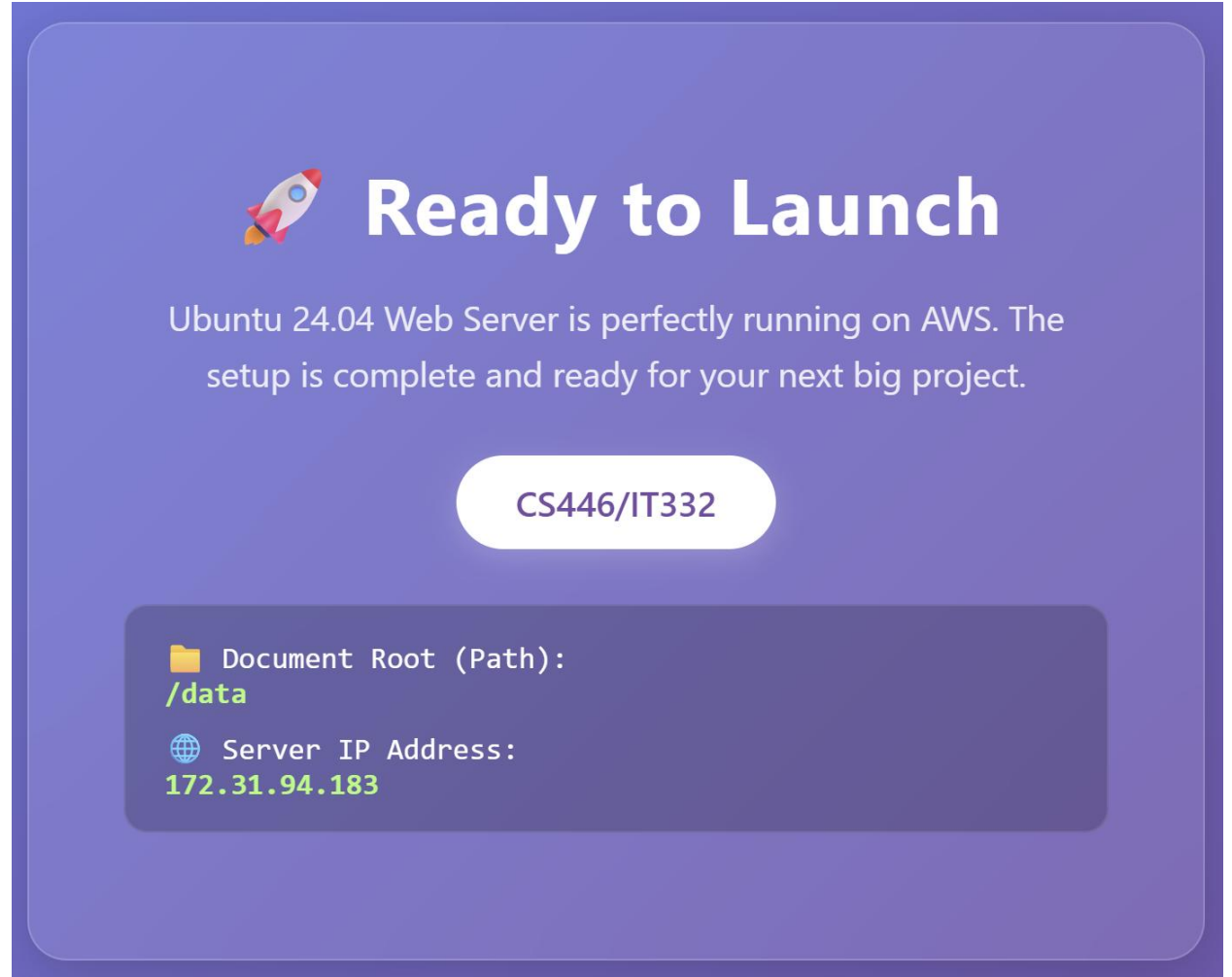
```
$ ls -l /data/index.html
```

```
ubuntu@ip-172-31-94-183:~$  
ubuntu@ip-172-31-94-183:~$ sudo cp /data2/index.html /data/index.html  
ubuntu@ip-172-31-94-183:~$ ls -l /data/index.html  
-rw-r--r-- 1 root root 2403 Sep  1 08:06 /data/index.html  
ubuntu@ip-172-31-94-183:~$  
ubuntu@ip-172-31-94-183:~$  
ubuntu@ip-172-31-94-183:~$  
ubuntu@ip-172-31-94-183:~$  
ubuntu@ip-172-31-94-183:~$
```

Check our web content from restored snapshot

- Access our web server

`http://<IP>:80`



 **Ready to Launch**

Ubuntu 24.04 Web Server is perfectly running on AWS. The setup is complete and ready for your next big project.

CS446/IT332

 Document Root (Path):
/data

 Server IP Address:
172.31.94.183

An aerial photograph of a terraced rice field landscape. A light-colored, winding river or path cuts through the terraced fields, which are filled with water and show varying shades of green and brown. The terraces are carved into the hillsides, creating a series of steps. The overall scene is a mix of natural and agricultural elements.

Clean Up Lab Resources

Remove instance

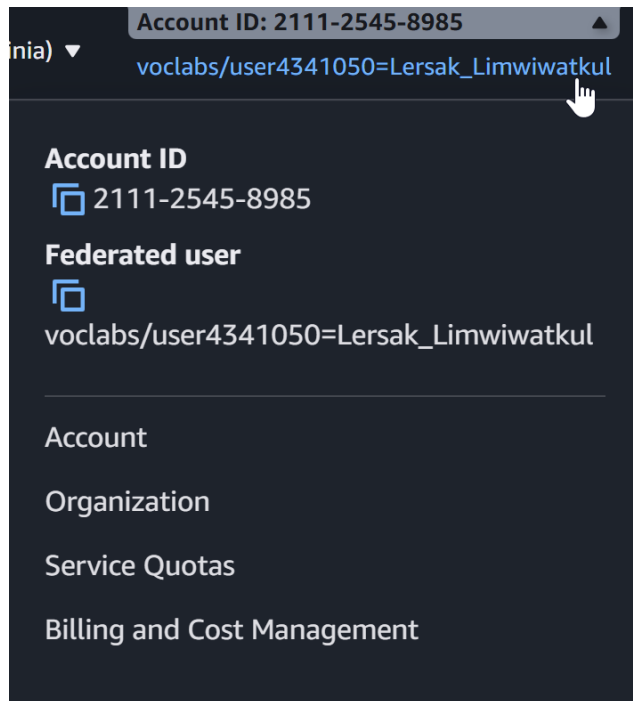
- Select any instance
- Instance state → Terminate (delete) instance

The screenshot shows the AWS Management Console 'Instances' page. At the top, it says 'Instances (1/1)' with an 'Info' link. Below this, there's a 'Last updated about 1 hour ago' message and a refresh icon. Action buttons include 'Connect', 'Instance state', 'Actions' (which is open), and 'Launch instances'. A 'Saved filter sets' section shows 'Choose filter set'. A search bar contains 'Find Instance'. The instance list has columns for selection, Name, Instance ID, and Instance type. One instance is listed: 'joe111111' with ID 'i-04c34722' and type 't3.small'. The 'Actions' dropdown menu is open, showing options: 'Stop instance', 'Start instance', 'Reboot instance', 'Hibernate instance', and 'Terminate (delete) instance' (which is highlighted by a mouse cursor).

	Name	Instance ID	Instance type
☒	joe111111	i-04c34722	t3.small

Billings and Cost Management

- Billing and Cost Management Home



Billing and Cost Management home [Info](#) [Provide feedback](#) [Need help?](#)

Cost summary [Info](#)

Month-to-date cost <u>\$0.11</u> - compared to last month for same period	Last month's cost for same time period <u>\$0.00</u> Aug 1 – 6
Total forecasted cost for current month <u>\$0.17</u> - compared to last month's total costs	Last month's total cost <u>\$0.00</u>

[View bill](#)

Sign out and End Lab

- Sign out from AWS Console

- Click End Lab

